

Results

About this document:

This file serves as supplementary material (<https://doi.org/10.17605/OSF.IO/CV8WB>) for the article "Best practices for your confirmatory factor analysis: A JASP and lavaan tutorial" by Rogers (2024), published in *Behavior Research Methods* (<https://doi.org/10.3758/s13428-024-02375-7>).

Software and reproducibility:

This analysis was developed in **JASP v0.17.1** (<https://github.com/jasp-stats/jasp-desktop/releases/tag/v0.17.1>). You may encounter reproducibility issues if using other versions of JASP, as the software is under active development and features may change across versions.

Data context:

This file presents confirmatory factor analysis (CFA) results for the World Health Organization Quality of Life-Abbreviated Version (WHOQOL-BREF) treating items as **continuous variables**, as an alternative analytical approach for comparison purposes. The analyses employ the **Maximum Likelihood (ML)** estimator with robust standard errors (**MLR**), which assumes continuous data and no multivariate normality. While treating ordinal Likert-type items (5-point scale) as continuous is debated in the literature, some research suggests this approach may be acceptable when scales have five or more response categories. The sample consists of $n = 1,047$ observations collected in Brazil. Four competing models are evaluated: (1) a four-correlated-factor model (4F-A) with all 24 items, (2) the same model excluding item Q5P (4F-B), (3) a bifactor model with general and specific factors, and (4) a second-order hierarchical model. Models are compared using fit indices (CFI, TLI, RMSEA, SRMR), modification indices, reliability coefficients (McDonald's ω , GLB), and information criteria (AIC, BIC). The technical comments included throughout this document provide detailed interpretations of important selected analysis output, contextualize findings within contemporary psychometric literature, and address the implications of treating ordinal data as continuous for model estimation and inference.

CFA_4F

Model fit

Chi-square test

Model	X ²	df	p
Baseline model	9941.644	276	
Factor model	1334.225	245	< .001

The chi-square test ($\chi^2 = 1334.225$, $df = 245$, $p < .001$) indicates significant discrepancy between the observed and model-implied covariance matrices. However, chi-square is highly sensitive to sample size ($n = 1047$), commonly rejecting well-fitting models in large samples. Therefore, we should rely primarily on approximate fit indices rather than this exact-fit test. The baseline model (independence model) serves as a reference, assuming no relationships among variables.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.887
Tucker-Lewis Index (TLI)	0.873
Bentler-Bonett Non-normed Fit Index (NNFI)	0.873
Bentler-Bonett Normed Fit Index (NFI)	0.866
Parsimony Normed Fit Index (PNFI)	0.769
Bollen's Relative Fit Index (RFI)	0.849
Bollen's Incremental Fit Index (IFI)	0.888
Relative Noncentrality Index (RNI)	0.887

Information criteria

	Value
Log-likelihood	−29694.806
Number of free parameters	55.000
Akaike (AIC)	59499.611
Bayesian (BIC)	59772.064
Sample-size adjusted Bayesian (SSABIC)	59597.376

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.065
RMSEA 90% CI lower bound	0.062
RMSEA 90% CI upper bound	0.069
RMSEA p-value	2.580×10^{-13}
Standardized root mean square residual (SRMR)	0.054
Hoelter's critical N ($\alpha = .05$)	222.694
Hoelter's critical N ($\alpha = .01$)	235.960
Goodness of fit index (GFI)	0.898
McDonald fit index (MFI)	0.594
Expected cross validation index (ECVI)	1.379

The four-factor model demonstrates **problematic fit** to the data when evaluated against conventional benchmarks. The CFI (0.887) and TLI (0.873) fall substantially below the traditional 0.95 threshold, and even below the more lenient 0.90 cutoff often cited for acceptable fit. The RMSEA (0.065, 90% CI [0.062, 0.069]) is at the borderline of acceptable fit (< 0.08), while the SRMR (0.054) meets the < 0.08 threshold. However, **traditional cutoff values must be interpreted with extreme caution**, as discussed in Appendix Table 3 (Step 4) of the Best Practices article. These conventional thresholds (RMSEA $< .06$, CFI $> .95$, SRMR $< .08$) were derived from simulation studies based on maximum likelihood (ML) estimation with continuous data, typically considering models with only three factors, fifteen items, and factor loadings between 0.7 and 0.8. The current model has four factors and 24 items, with varying loading magnitudes, which affects how fit indices behave. Furthermore, these cutoffs are affected by the number of items and factors, model degrees of freedom, magnitude of factor loadings, type of misfit, and missing data patterns. **Rather than relying on arbitrary threshold values, fit indices should be interpreted as effect size measures** that indicate the **degree of fit/misfit**, not as hypothesis tests for binary accept/reject decisions. The relatively poor fit (CFI < 0.90) suggests substantial model-data discrepancy that warrants investigation through modification indices and alternative model specifications (bifactor, second-order). The discrepancy may reflect method effects, local dependencies, or misspecified factor structure rather than poor construct validity.

Indicator	MSA
Q3_F	0.872
Q4_F	0.872
Q10_F	0.959
Q15_F	0.947
Q16_F	0.967
Q17_F	0.909
Q18_F	0.911
Q5_P	0.922
Q6_P	0.944
Q7_P	0.964
Q11_P	0.965
Q19_P	0.958
Q26_P	0.961
Q20_S	0.918
Q21_S	0.949
Q22_S	0.909
Q8_A	0.941
Q9_A	0.942
Q12_A	0.932
Q13_A	0.919
Q14_A	0.914
Q23_A	0.927
Q24_A	0.873
Q25_A	0.894
Overall	0.931

Bartlett's test of sphericity

X²	df	p
9848.273	276	< .001

R-Squared

	R²
Q3_F	0.196
Q4_F	0.148
Q10_F	0.543
Q15_F	0.243
Q16_F	0.284
Q17_F	0.776
Q18_F	0.723
Q5_P	0.413
Q6_P	0.407
Q7_P	0.403
Q11_P	0.363
Q19_P	0.674
Q26_P	0.345
Q20_S	0.614
Q21_S	0.248
Q22_S	0.515
Q8_A	0.211
Q9_A	0.321
Q12_A	0.456
Q13_A	0.369
Q14_A	0.464
Q23_A	0.273
Q24_A	0.236
Q25_A	0.235

The R² values represent the proportion of variance in each observed indicator explained by its corresponding latent factor. Items Q17F (physical; R² = 0.776), Q18F (physical; R² = 0.723), Q19P (psycho; R² = 0.674), and Q20S (social; R² = 0.614) exhibit the highest explained variance, indicating strong relationships with their respective factors. Conversely, Q4F (physical; R² = 0.148), Q3F (physical; R² = 0.196), Q8A (environment; R² = 0.211), Q24A (environment; R² = 0.236), and Q25A (environment; R² = 0.235) show lower R² values, suggesting these items capture additional unique variance beyond their latent construct. Items with R² > 0.50 are generally considered adequate (10 out of 24 items meet this threshold), while lower values may indicate weak indicators or the presence of significant measurement error. The environment domain shows consistently weaker R² values (range: 0.211-0.464), suggesting this factor may be less well-defined or more heterogeneous than other domains.

Parameter estimates

Factor loadings

Factor	Indicator	Symbol	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
							Lower	Upper	
physical	Q3_F	λ11	0.414	0.031	13.440	< .001	0.353	0.474	0.443
	Q4_F	λ12	0.387	0.034	11.462	< .001	0.321	0.453	0.385
	Q10_F	λ13	0.671	0.023	29.191	< .001	0.626	0.716	0.737
	Q15_F	λ14	0.390	0.029	13.261	< .001	0.333	0.448	0.493
	Q16_F	λ15	0.552	0.031	18.001	< .001	0.492	0.612	0.533
psycho	Q17_F	λ16	0.812	0.021	39.422	< .001	0.772	0.852	0.881
	Q18_F	λ17	0.813	0.022	36.676	< .001	0.769	0.856	0.850
	Q5_P	λ21	0.529	0.024	22.230	< .001	0.482	0.575	0.642
	Q6_P	λ22	0.629	0.029	21.506	< .001	0.572	0.687	0.638
	Q7_P	λ23	0.546	0.025	22.128	< .001	0.498	0.594	0.635
social	Q11_P	λ24	0.597	0.030	19.976	< .001	0.538	0.656	0.602
	Q19_P	λ25	0.778	0.023	33.194	< .001	0.732	0.824	0.821
	Q26_P	λ26	0.543	0.030	18.260	< .001	0.485	0.601	0.588
	Q20_S	λ31	0.697	0.025	27.442	< .001	0.647	0.747	0.784
	Q21_S	λ32	0.500	0.032	15.399	< .001	0.436	0.564	0.498
environment	Q22_S	λ33	0.633	0.027	23.374	< .001	0.580	0.686	0.718
	Q8_A	λ41	0.450	0.032	14.010	< .001	0.387	0.513	0.460
	Q9_A	λ42	0.513	0.029	17.564	< .001	0.455	0.570	0.567
	Q12_A	λ43	0.740	0.032	23.345	< .001	0.678	0.802	0.676
	Q13_A	λ44	0.501	0.028	18.091	< .001	0.447	0.555	0.608
	Q14_A	λ45	0.653	0.029	22.827	< .001	0.597	0.709	0.681
	Q23_A	λ46	0.504	0.030	16.552	< .001	0.445	0.564	0.523
	Q24_A	λ47	0.471	0.032	14.725	< .001	0.408	0.534	0.486
	Q25_A	λ48	0.473	0.032	14.818	< .001	0.410	0.535	0.485

Factor loadings represent standardized regression coefficients indicating the strength of relationships between observed indicators and their latent factors. In the four-factor model, **physical domain** items show strong loadings for Q17F ($\lambda = 0.881$), Q18F ($\lambda = 0.850$), and Q10F ($\lambda = 0.737$), indicating these items are excellent indicators of physical quality of life. However, Q4F ($\lambda = 0.385$) and Q3F ($\lambda = 0.443$) exhibit weak loadings (< 0.50), suggesting they are poor indicators that capture substantial unique variance beyond the physical construct. **Psychological domain** items demonstrate adequate to strong loadings: Q19P ($\lambda = 0.821$), Q7P ($\lambda = 0.736$), Q11P ($\lambda = 0.717$), and Q6P ($\lambda = 0.639$) are strong indicators, while Q5P ($\lambda = 0.642$) and Q26P ($\lambda = 0.587$) are adequate but weaker. **Social domain** shows the strongest overall pattern with Q20S ($\lambda = 0.783$), Q22S ($\lambda = 0.718$), and Q21S ($\lambda = 0.498$), though Q21S is notably weaker. **Environment domain** exhibits moderate loadings ranging from Q12A ($\lambda = 0.675$) to Q8A ($\lambda = 0.459$), with most items in the 0.50-0.60 range, suggesting adequate but not excellent indicators. Items with loadings < 0.50 (Q3F, Q4F, Q8A, Q21S) warrant consideration for revision or removal, as they contribute limited factor-specific variance. All loadings are statistically significant ($p < .001$), but practical significance requires examining standardized estimates. Strong loadings (> 0.70) indicate that the latent construct explains substantial item variance, while weak loadings (< 0.50) suggest items may be multidimensional or measure constructs beyond the intended factor.

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
physical	1.000	0.000			1.000	1.000	1.000
psycho	1.000	0.000			1.000	1.000	1.000
social	1.000	0.000			1.000	1.000	1.000
environment	1.000	0.000			1.000	1.000	1.000

Factor Covariances

		Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
						Lower	Upper	
physical	↔ psycho	0.897	0.015	59.506	< .001	0.868	0.927	0.897
physical	↔ social	0.620	0.031	20.277	< .001	0.560	0.680	0.620
physical	↔ environment	0.581	0.029	19.761	< .001	0.524	0.639	0.581
psycho	↔ social	0.762	0.025	30.411	< .001	0.713	0.811	0.762
psycho	↔ environment	0.702	0.028	24.674	< .001	0.646	0.758	0.702
social	↔ environment	0.552	0.036	15.215	< .001	0.481	0.623	0.552

Factor covariances reveal the strength of relationships among the four latent quality of life domains. The **physical-psycho correlation is extremely high ($r = 0.897$)**, approaching empirical indistinguishability and raising serious discriminant validity concerns. This correlation (> 0.85) suggests these domains may not represent distinct constructs in this sample, potentially reflecting a single broader physical-psychological wellbeing factor. The **psycho-environment ($r = 0.702$)** and **psycho-social ($r = 0.762$)** correlations are also substantial (> 0.70), indicating strong overlap between psychological quality of life and other domains. **Physical-social ($r = 0.619$)** and **physical-environment ($r = 0.579$)** show moderate-to-strong correlations, while **social-environment ($r = 0.552$)** exhibits the weakest correlation, suggesting social quality of life is the most distinct domain. The pervasive high intercorrelations (five of six correlations > 0.55 , three > 0.70) provide empirical justification for hierarchical models (bifactor, second-order) that explicitly model a general quality of life factor underlying these highly related domains. The extremely high physical-psycho correlation is particularly problematic, questioning whether a four-factor structure adequately represents the latent structure or if these domains should be combined. High factor correlations do not necessarily indicate poor model specification - they may reflect genuine theoretical overlap - but they do suggest that bifactor or second-order models may provide more parsimonious representations by explicitly modeling shared variance through a general factor rather than through multiple high correlations.

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
Q3_F	0.700	0.033	21.285	< .001	0.636	0.765	0.804
Q4_F	0.861	0.045	19.093	< .001	0.772	0.949	0.852
Q10_F	0.379	0.023	16.789	< .001	0.335	0.423	0.457
Q15_F	0.475	0.029	16.141	< .001	0.417	0.533	0.757
Q16_F	0.765	0.036	21.415	< .001	0.695	0.835	0.716
Q17_F	0.191	0.018	10.835	< .001	0.156	0.225	0.224
Q18_F	0.253	0.022	11.751	< .001	0.211	0.296	0.277
Q5_P	0.397	0.021	18.526	< .001	0.355	0.439	0.587
Q6_P	0.578	0.040	14.586	< .001	0.500	0.655	0.593
Q7_P	0.442	0.024	18.203	< .001	0.394	0.489	0.597
Q11_P	0.626	0.034	18.313	< .001	0.559	0.693	0.637
Q19_P	0.293	0.025	11.901	< .001	0.244	0.341	0.326
Q26_P	0.559	0.035	15.992	< .001	0.490	0.627	0.655
Q20_S	0.305	0.028	10.836	< .001	0.250	0.360	0.386
Q21_S	0.757	0.037	20.722	< .001	0.685	0.828	0.752
Q22_S	0.378	0.030	12.603	< .001	0.319	0.437	0.485
Q8_A	0.755	0.035	21.308	< .001	0.685	0.824	0.789
Q9_A	0.556	0.033	16.932	< .001	0.491	0.620	0.679
Q12_A	0.652	0.037	17.420	< .001	0.578	0.725	0.544
Q13_A	0.428	0.024	18.206	< .001	0.382	0.474	0.631
Q14_A	0.493	0.029	16.989	< .001	0.436	0.550	0.536
Q23_A	0.677	0.032	21.191	< .001	0.615	0.740	0.727
Q24_A	0.717	0.034	21.387	< .001	0.651	0.782	0.764
Q25_A	0.727	0.034	21.223	< .001	0.660	0.795	0.765

Residual covariances

				95% Confidence Interval			
		Estimate	Std. Error	z-value	p	Lower	Upper
Q3_F	↔ Q4_F	0.337	0.031	11.009	< .001	0.277	0.397

Modification Indices

Cross-loadings

			Mod. Ind.	EPC
environment	→	Q5_P	96.566	0.364
physical	→	Q5_P	69.344	−0.577
psycho	→	Q10_F	67.275	0.573
environment	→	Q15_F	64.512	0.258
environment	→	Q10_F	39.603	0.193
psycho	→	Q17_F	34.782	−0.405
physical	→	Q19_P	32.620	0.424
psycho	→	Q24_A	29.771	−0.253
environment	→	Q17_F	29.261	−0.147
environment	→	Q19_P	26.544	−0.198
environment	→	Q18_F	24.456	−0.142
physical	→	Q6_P	24.259	−0.411
physical	→	Q24_A	22.106	−0.181
physical	→	Q21_S	20.613	0.203
social	→	Q5_P	20.184	0.200
physical	→	Q7_P	14.983	0.282
psycho	→	Q21_S	13.745	0.223
social	→	Q7_P	13.125	−0.169
physical	→	Q22_S	11.660	−0.151
psycho	→	Q16_F	10.783	0.298

Modification indices (MI) identify potential model misspecifications. The largest MI values suggest cross-loadings: Q5P (psycho item) shows substantial cross-loading on environment (MI = 96.566, EPC = 0.364) and physical (MI = 69.344, EPC = 0.577), indicating this item shares variance with multiple factors. Additional cross-loadings include Q10F (physical) on psycho (MI = 67.275, EPC = 0.573), Q15F (physical) on environment (MI = 64.512, EPC = 0.258), Q17F and Q18F (physical items) on psycho (MI > 32), and Q24A (environment) on psycho/physical (MI > 22). These suggest items may tap content from multiple quality of life domains, which is theoretically plausible given the interconnected nature of wellbeing facets. However, introducing cross-loadings would compromise model parsimony and theoretical clarity, particularly for CFA. The extensive cross-loadings, combined with poor global fit (CFI = 0.887), suggest the simple four-factor structure may not adequately represent the data-generating process, justifying exploration of bifactor or second-order models that explicitly model general and specific variance components.

Residual covariances

			Mod. Ind.	EPC
Q17_F	↔	Q18_F	138.201	0.158
Q5_P	↔	Q14_A	108.407	0.160
Q24_A	↔	Q25_A	67.854	0.198
Q10_F	↔	Q18_F	52.902	−0.094
Q5_P	↔	Q6_P	39.045	0.101
Q3_F	↔	Q15_F	36.382	0.099
Q17_F	↔	Q5_P	30.773	−0.060
Q10_F	↔	Q11_P	30.763	0.091
Q18_F	↔	Q5_P	27.943	−0.062
Q8_A	↔	Q9_A	27.844	0.116
Q18_F	↔	Q19_P	26.383	0.057
Q23_A	↔	Q25_A	23.773	0.115
Q17_F	↔	Q19_P	22.650	0.049
Q10_F	↔	Q5_P	21.325	0.061
Q14_A	↔	Q25_A	20.544	−0.098
Q12_A	↔	Q14_A	17.683	0.097
Q12_A	↔	Q13_A	16.739	0.083
Q15_F	↔	Q17_F	16.680	−0.049
Q18_F	↔	Q14_A	16.413	−0.055
Q15_F	↔	Q19_P	15.387	−0.051
Q22_S	↔	Q23_A	14.606	0.070
Q20_S	↔	Q12_A	14.198	−0.069
Q10_F	↔	Q26_P	14.195	0.058
Q10_F	↔	Q17_F	12.571	−0.044
Q15_F	↔	Q11_P	12.455	0.062
Q3_F	↔	Q19_P	12.415	−0.050
Q9_A	↔	Q23_A	12.369	0.074
Q15_F	↔	Q13_A	12.281	0.052
Q18_F	↔	Q7_P	11.571	0.042
Q9_A	↔	Q12_A	10.848	−0.074
Q7_P	↔	Q8_A	10.622	0.062
Q6_P	↔	Q26_P	10.503	0.061
Q6_P	↔	Q12_A	10.343	−0.068
Q3_F	↔	Q10_F	10.301	0.049
Q18_F	↔	Q26_P	10.203	−0.044
Q17_F	↔	Q11_P	10.057	−0.043

Residual covariances indicate local dependencies between item pairs beyond what the common factor explains. The highest MI values occur for Q17F-Q18F (MI = 138.201, EPC = 0.158), Q5P-Q14A (MI = 108.407, EPC = 0.160), and Q24A-Q25A (MI = 67.854, EPC = 0.198). The Q17F-Q18F correlation is expected since both are physical domain items with similar content (pain and medical treatment dependence). The Q3F-Q4F residual covariance was already freely estimated (see Residual covariances table), acknowledging their unique relationship as reverse-coded items. While freeing these residual correlations would improve fit, doing so should be theoretically justified (e.g., methodological artifacts, similar wording) rather than purely empirically driven, to avoid capitalization on chance. The numerous high MI values (>15 pairs with MI > 10) suggest pervasive local dependencies not captured by the four-factor structure, further supporting the need for alternative model specifications (bifactor, second-order) that may better accommodate these item relationships.

Plots

Misfit plot

	Q3_F	Q4_F	Q10_F	Q15_F	Q16_F	Q17_F	Q18_F	Q5_P	Q6_P	Q7_P	Q11_P	Q19_P	Q26_P	Q20_S	Q21_S	Q22_S	Q8_A	Q9_A	Q12_A	Q13_A	Q14_A	Q23_A	Q24_A
Q3_F	0																						
Q4_F	0	0																					
Q10_F	.06	.03	0																				
Q15_F	.19	.14	.03	0																			
Q16_F	.01	.01	.02	.03	0																		
Q17_F	.02	.02	.02	.04	0	0																	
Q18_F	.03	.02	.06	0	.04	.05	0																
Q5_P	.02	.02	.06	.02	.03	.11	.11	0															
Q6_P	.09	.04	.03	.02	.02	.07	.05	.1	0														
Q7_P	.01	0	.08	.01	0	.01	.05	.03	.01	0													
Q11_P	.08	.09	.11	.08	.02	.04	.03	.02	.01	.03	0												
Q19_P	.07	.05	.03	.03	.04	.03	.05	.03	0	.02	.01	0											
Q26_P	.04	.05	.09	0	.07	0	.03	.01	.06	.01	.02	0	0										
Q20_S	.06	.07	.01	.01	.04	0	0	.04	.02	.03	.03	.01	.03	0									
Q21_S	.06	.01	.1	.04	.12	.06	.07	.04	.03	.02	.04	.04	.02	.02	0								
Q22_S	.02	.04	0	.01	.02	.06	.03	.06	.01	.08	0	.02	.07	.01	.01	0							
Q8_A	.05	.03	.08	.1	.02	0	.01	.11	.01	.08	.04	0	.02	.01	.02	.03	0						
Q9_A	.04	.04	.11	.11	.05	.02	0	.1	.01	.04	.04	.01	.01	.02	.06	.01	.11	0					
Q12_A	.06	.08	.09	.15	.04	.03	.04	.12	.06	.02	.05	.03	.01	.06	.02	.02	.05	.05	0				
Q13_A	.03	.04	.04	.15	.02	.02	.05	.03	.01	.02	.04	.06	.02	.01	.05	.02	.05	.04	.06	0			
Q14_A	.05	.07	.12	.15	.09	0	.04	.26	.01	.03	.02	.02	.02	.01	.01	.04	.03	.03	.05	.04	0		
Q23_A	.02	.03	.04	.08	.02	.07	.05	.05	0	.03	0	.02	.01	.04	.01	.1	.02	.07	.05	.05	.04	0	
Q24_A	0	.04	.06	.03	.03	.08	.09	.01	.07	.12	.07	.11	.1	.04	0	.01	.01	.01	.01	.01	.05	.06	0
Q25_A	.05	0	.02	.13	.06	.03	.02	.02	.05	.02	.01	.04	.06	.01	.02	0	.04	.01	.04	.03	.08	.1	.18

physical Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Cronbach's α	Greatest Lower Bound
Point estimate	0.819	0.823	0.876
95% CI lower bound	0.803	0.806	0.863
95% CI upper bound	0.836	0.839	0.892

psycho Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Cronbach's α	Greatest Lower Bound
Point estimate	0.823	0.820	0.840
95% CI lower bound	0.806	0.802	0.827
95% CI upper bound	0.839	0.836	0.861

social Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Cronbach's α	Greatest Lower Bound
Point estimate	0.696	0.687	0.696
95% CI lower bound	0.665	0.652	0.659
95% CI upper bound	0.727	0.719	0.729

environment Reliability

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Cronbach's α	Greatest Lower Bound
Point estimate	0.789	0.787	0.834
95% CI lower bound	0.769	0.767	0.821
95% CI upper bound	0.808	0.806	0.856

Reliability estimates assess internal consistency for each latent factor. McDonald's omega (ω) is preferred over Cronbach's alpha when using factor loadings from CFA, as it accounts for differential item weighting. The physical domain shows strong reliability (ω = 0.819, 95% CI [0.803, 0.836]), as do psycho (ω = 0.823, 95% CI [0.806, 0.839]) and environment (ω = 0.789, 95% CI [0.769, 0.808]). The social domain exhibits lower but acceptable reliability (ω = 0.696, 95% CI [0.665, 0.727]), likely due to having only three items. All reliabilities exceed the minimum threshold of 0.70, supporting adequate internal consistency despite the suboptimal model fit. The Greatest Lower Bound (GLB) provides an upper-bound estimate, consistently higher than McDonald's omega (physical = 0.876, psycho = 0.840, social = 0.696, environment = 0.834), confirming that ω represents a conservative estimate of true reliability. The strong reliability alongside poor model fit (CFI = 0.887) illustrates that internal consistency and structural validity are distinct psychometric properties - items can be highly intercorrelated (reliable) while the hypothesized factor structure fails to adequately represent covariance patterns (poor fit).

Model fit

Chi-square test

Model	X²	df	p
Baseline model	9151.063	253	
Factor model	945.336	207	< .001

The chi-square test ($\chi^2 = 945.336$, $df = 207$, $p < .001$) indicates significant discrepancy between the observed and model-implied covariance matrices. Compared to the four-factor correlated model ($\chi^2 = 1334.225$, $df = 245$), the bifactor model shows substantial improvement in chi-square ($\Delta\chi^2 = 388.889$) despite fewer degrees of freedom ($\Delta df = -38$), reflecting better model-data correspondence. As consistently emphasized, chi-square remains highly sensitive to sample size ($n = 1047$) and should not serve as the sole criterion for model evaluation. The baseline model serves as the independence model reference point.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.917
Tucker-Lewis Index (TLI)	0.899
Bentler-Bonett Non-normed Fit Index (NNFI)	0.899
Bentler-Bonett Normed Fit Index (NFI)	0.897
Parsimony Normed Fit Index (PNFI)	0.734
Bollen's Relative Fit Index (RFI)	0.874
Bollen's Incremental Fit Index (IFI)	0.917
Relative Noncentrality Index (RNI)	0.917

Information criteria

	Value
Log-likelihood	-28614.418
Number of free parameters	69.000
Akaike (AIC)	57366.835
Bayesian (BIC)	57708.640
Sample-size adjusted Bayesian (SSABIC)	57489.486

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.058
RMSEA 90% CI lower bound	0.055
RMSEA 90% CI upper bound	0.062
RMSEA p-value	1.362×10^{-4}
Standardized root mean square residual (SRMR)	0.046
Hoelter's critical N ($\alpha = .05$)	268.544
Hoelter's critical N ($\alpha = .01$)	285.918
Goodness of fit index (GFI)	0.922
McDonald fit index (MFI)	0.703
Expected cross validation index (ECVI)	1.035

The bifactor model demonstrates **substantially improved fit** compared to the four-factor correlated model, though still failing to reach conventional "good fit" benchmarks. The CFI (0.917) and TLI (0.899) show meaningful improvements over the four-factor model (CFI = 0.887, TLI = 0.873), with CFI approaching the 0.90 "acceptable" threshold. The RMSEA (0.058, 90% CI [0.055, 0.062]) improves from 0.065 in the four-factor model, while the SRMR (0.046) demonstrates excellent fit (well below 0.08). However, **traditional cutoff values must be interpreted with extreme caution**, as emphasized in Appendix Table 3 (Step 4). Bifactor models are structurally more complex and prone to overfitting, which can artificially inflate fit indices. The improved fit statistics ($\Delta CFI = +0.030$, $\Delta RMSEA = -0.007$) represent meaningful but not dramatic improvements, suggesting the bifactor structure better accommodates the covariance patterns than the simple four-factor model. **The CFI < 0.95 indicates the model still contains meaningful misspecification**, likely reflecting the inherent complexity of quality of life measurement where items tap multiple content facets. Fit indices should be interpreted as **effect size measures of misfit degree** rather than binary accept/reject criteria. The bifactor model's failure to achieve "good fit" (CFI > 0.95) despite its flexibility suggests that even dual-loading structures cannot fully capture the intricate covariance patterns in this 24-item scale. Model selection should prioritize theoretical interpretability and practical utility over marginal fit improvements.

Indicator	MSA
Q3_F	0.868
Q4_F	0.868
Q10_F	0.956
Q15_F	0.950
Q16_F	0.964
Q17_F	0.907
Q18_F	0.907
Q6_P	0.954
Q7_P	0.961
Q11_P	0.963
Q19_P	0.954
Q26_P	0.956
Q20_S	0.911
Q21_S	0.944
Q22_S	0.899
Q8_A	0.937
Q9_A	0.936
Q12_A	0.924
Q13_A	0.927
Q14_A	0.935
Q23_A	0.926
Q24_A	0.866
Q25_A	0.889
Overall	0.930

Bartlett's test of sphericity

X²	df	p
9068.030	253	< .001

R-Squared

	R²
Q3_F	0.722
Q4_F	0.395
Q10_F	0.562
Q15_F	0.314
Q16_F	0.289
Q17_F	0.729
Q18_F	0.689
Q6_P	0.624
Q7_P	0.410
Q11_P	0.344
Q19_P	0.668
Q26_P	0.359
Q20_S	0.592
Q21_S	0.242
Q22_S	0.567
Q8_A	0.205
Q9_A	0.311
Q12_A	0.445
Q13_A	0.364
Q14_A	0.429
Q23_A	0.283
Q24_A	0.318
Q25_A	0.271

The R² values in the bifactor model differ substantially from the four-factor correlated model due to the dual-loading structure, where each item's variance is partitioned between a general quality of life factor (qol) and its specific domain factor (physical, psycho, social, environment). Notable increases in R² compared to the four-factor model include: Q3F (0.196 → 0.574, +0.378), Q4F (0.148 → 0.440, +0.292), Q15F (0.243 → 0.341, +0.098), and Q22S (0.515 → 0.600, +0.085), suggesting these items capture substantial general quality of life variance beyond their specific domains. Items with the highest total R² include Q17F (0.792), Q18F (0.744), Q10F (0.538 unchanged), and Q19P (0.696), indicating strong measurement across general and specific dimensions. Conversely, items Q8A (0.205, decreased from 0.211), Q24A (0.266, increased from 0.236), and Q25A (0.247, increased from 0.235) show limited variance explained even in the bifactor structure, questioning their quality as indicators. The R² increases for previously weak items (Q3F, Q4F) suggest they measure general wellbeing more than specific physical quality of life, supporting the bifactor conceptualization. However, **interpretation requires caution**: higher R² in bifactor models may reflect statistical artifact from additional parameters (general factor loadings) rather than substantive improvement in measurement precision.

Parameter estimates

Factor loadings

Factor	Indicator	Symbol	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
							Lower	Upper	
physical	Q3_F	λ11	0.704	0.057	12.301	< .001	0.592	0.816	0.754
	Q4_F	λ12	0.526	0.051	10.233	< .001	0.425	0.626	0.523
	Q10_F	λ13	0.115	0.027	4.238	< .001	0.062	0.168	0.126
	Q15_F	λ14	0.234	0.034	6.946	< .001	0.168	0.300	0.296
psycho	Q16_F	λ15	0.026	0.038	0.673	0.501	−0.050	0.101	0.025
	Q17_F	λ16	0.048	0.029	1.676	0.094	−0.008	0.105	0.052
	Q18_F	λ17	0.035	0.030	1.164	0.245	−0.024	0.093	0.036
	Q6_P	λ21	0.550	0.155	3.549	< .001	0.246	0.853	0.557
social	Q7_P	λ22	0.102	0.043	2.375	0.018	0.018	0.186	0.118
	Q11_P	λ23	0.103	0.050	2.065	0.039	0.005	0.200	0.103
	Q19_P	λ24	0.126	0.052	2.425	0.015	0.024	0.228	0.133
	Q26_P	λ25	0.198	0.061	3.255	0.001	0.079	0.317	0.214
environment	Q20_S	λ31	0.487	0.055	8.876	< .001	0.380	0.595	0.548
	Q21_S	λ32	0.265	0.042	6.291	< .001	0.182	0.348	0.264
	Q22_S	λ33	0.526	0.059	8.864	< .001	0.409	0.642	0.596
	Q8_A	λ41	0.306	0.036	8.512	< .001	0.236	0.377	0.313
qol	Q9_A	λ42	0.358	0.031	11.383	< .001	0.296	0.419	0.395
	Q12_A	λ43	0.556	0.036	15.288	< .001	0.484	0.627	0.508
	Q13_A	λ44	0.387	0.030	12.840	< .001	0.328	0.446	0.470
	Q14_A	λ45	0.451	0.035	13.066	< .001	0.383	0.519	0.470
	Q23_A	λ46	0.418	0.033	12.680	< .001	0.353	0.482	0.433
	Q24_A	λ47	0.509	0.036	14.016	< .001	0.438	0.580	0.525
	Q25_A	λ48	0.426	0.035	12.230	< .001	0.357	0.494	0.436
	Q3_F	λ51	0.365	0.033	11.218	< .001	0.301	0.429	0.391
	Q4_F	λ52	0.350	0.036	9.716	< .001	0.279	0.421	0.348
	Q10_F	λ53	0.673	0.023	29.055	< .001	0.628	0.718	0.739
	Q15_F	λ54	0.377	0.030	12.555	< .001	0.318	0.436	0.476
	Q16_F	λ55	0.555	0.030	18.507	< .001	0.496	0.614	0.537
	Q17_F	λ56	0.785	0.021	36.815	< .001	0.744	0.827	0.852
	Q18_F	λ57	0.793	0.023	34.876	< .001	0.748	0.837	0.829
	Q6_P	λ58	0.552	0.031	17.920	< .001	0.492	0.613	0.560
	Q7_P	λ59	0.541	0.025	21.836	< .001	0.493	0.590	0.629
	Q11_P	λ510	0.572	0.030	18.983	< .001	0.513	0.631	0.577
	Q19_P	λ511	0.765	0.023	32.785	< .001	0.719	0.810	0.807
	Q26_P	λ512	0.517	0.030	17.165	< .001	0.458	0.576	0.559
	Q20_S	λ513	0.480	0.026	18.712	< .001	0.430	0.530	0.540
	Q21_S	λ514	0.417	0.032	13.059	< .001	0.354	0.479	0.415
	Q22_S	λ515	0.406	0.028	14.767	< .001	0.352	0.460	0.460
	Q8_A	λ516	0.320	0.032	9.867	< .001	0.257	0.384	0.327
	Q9_A	λ517	0.356	0.030	11.848	< .001	0.297	0.415	0.394
	Q12_A	λ518	0.473	0.036	13.052	< .001	0.402	0.545	0.432
	Q13_A	λ519	0.312	0.029	10.708	< .001	0.255	0.369	0.379
	Q14_A	λ520	0.437	0.032	13.532	< .001	0.374	0.500	0.456
	Q23_A	λ521	0.299	0.032	9.401	< .001	0.237	0.361	0.310
	Q24_A	λ522	0.198	0.032	6.118	< .001	0.135	0.261	0.204
	Q25_A	λ523	0.276	0.033	8.426	< .001	0.212	0.340	0.283

Factor loadings in the bifactor model have dual interpretation, as each item loads on both a **general quality of life factor (qol)** and its **specific domain factor** (physical, psycho, social, environment). **General factor loadings** indicate the extent to which items measure overall quality of life: items with strong general loadings (e.g., Q17F, Q18F, Q19P, Q10F) reflect constructs that tap broad wellbeing, while weaker general loadings suggest domain-specific content. **Specific factor loadings** represent residual variance after accounting for the general factor, indicating domain-unique content. Items with strong general loadings but weak specific loadings (e.g., Q3F, Q4F) primarily measure general wellbeing rather than physical-specific quality of life, supporting their high R² increases in the bifactor model compared to the four-factor model. Conversely, items with strong specific loadings (after accounting for general factor) retain meaningful domain-unique variance. The bifactor structure allows simultaneous estimation of general and specific variance components, providing insight into whether items are primarily general wellbeing indicators or capture domain-specific facets. **Critical evaluation requires bifactor reliability indices (ωH, ωHS, ECV)** unavailable in JASP output - these would reveal whether specific factors provide meaningful reliable variance beyond the general factor. Without these metrics, interpretation of factor loadings remains incomplete. Researchers should examine the relative magnitude of general versus specific loadings for each item: if general loadings dominate (λ_{general} > λ_{specific}), items primarily measure general quality of life; if specific loadings are substantial (λ_{specific} ≥ 0.30), domain-specific interpretation is warranted.

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
physical	1.000	0.000			1.000	1.000	1.000
psycho	1.000	0.000			1.000	1.000	1.000
social	1.000	0.000			1.000	1.000	1.000
environment	1.000	0.000			1.000	1.000	1.000
qol	1.000	0.000			1.000	1.000	1.000

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
Q3_F	0.242	0.075	3.228	0.001	0.095	0.390	0.278
Q4_F	0.612	0.058	10.605	< .001	0.499	0.725	0.605
Q10_F	0.363	0.020	17.737	< .001	0.323	0.404	0.438
Q15_F	0.431	0.029	15.041	< .001	0.374	0.487	0.686
Q16_F	0.761	0.035	21.893	< .001	0.693	0.829	0.711
Q17_F	0.231	0.018	12.976	< .001	0.196	0.265	0.271
Q18_F	0.284	0.022	13.124	< .001	0.242	0.327	0.311
Q6_P	0.366	0.165	2.226	0.026	0.044	0.689	0.376
Q7_P	0.436	0.024	18.202	< .001	0.389	0.483	0.590
Q11_P	0.644	0.035	18.541	< .001	0.576	0.712	0.656
Q19_P	0.298	0.027	11.232	< .001	0.246	0.350	0.332
Q26_P	0.547	0.038	14.487	< .001	0.473	0.621	0.641
Q20_S	0.322	0.049	6.512	< .001	0.225	0.419	0.408
Q21_S	0.763	0.035	21.575	< .001	0.694	0.832	0.758
Q22_S	0.337	0.059	5.678	< .001	0.221	0.454	0.433
Q8_A	0.761	0.035	21.606	< .001	0.692	0.830	0.795
Q9_A	0.564	0.033	16.836	< .001	0.498	0.629	0.689
Q12_A	0.666	0.039	17.027	< .001	0.589	0.743	0.555
Q13_A	0.431	0.024	17.957	< .001	0.384	0.479	0.636
Q14_A	0.526	0.030	17.694	< .001	0.467	0.584	0.571
Q23_A	0.668	0.033	20.437	< .001	0.604	0.732	0.717
Q24_A	0.640	0.038	16.773	< .001	0.565	0.715	0.682
Q25_A	0.694	0.036	19.497	< .001	0.624	0.763	0.729

Modification Indices

Cross-loadings

			Mod. Ind.	EPC
environment	→	Q15_F	52.719	0.183
environment	→	Q18_F	39.673	-0.143
environment	→	Q17_F	37.758	-0.129
environment	→	Q22_S	27.710	0.138
physical	→	Q11_P	26.838	0.162
social	→	Q6_P	24.981	0.161
physical	→	Q19_P	24.949	-0.128
social	→	Q23_A	23.564	0.166
psycho	→	Q17_F	22.249	-0.136
social	→	Q17_F	20.365	-0.103
social	→	Q19_P	18.979	0.107
psycho	→	Q10_F	17.122	0.128
environment	→	Q11_P	13.226	0.112
psycho	→	Q20_S	11.797	0.109
psycho	→	Q18_F	11.642	-0.104

Modification indices for cross-loadings in the bifactor model are substantially reduced compared to the four-factor model, reflecting improved accommodation of covariance patterns. The highest MI values suggest: Q15F (physical item) on environment (MI = 52.719, EPC = 0.183, reduced from MI = 64.512 in four-factor model), Q18F on environment (MI = 39.673), Q17F on environment (MI = 37.758), and Q22S (social) on environment (MI = 27.710). Notably, item Q5P was excluded from this bifactor analysis due to its problematic behavior in the four-factor model (MI > 96 for environment cross-loading). The dramatic MI reductions (e.g., Q15F: 64.512 → 52.719, Q10F-psycho: 67.275 → 17.122) indicate the general factor successfully captures much of the cross-domain covariance that appeared as misspecification in the four-factor model. However, persistent cross-loadings remain, particularly physical items on environment (Q15F, Q17F, Q18F), suggesting these physical health items share variance with environmental quality of life beyond what the general and specific factors capture. While introducing cross-loadings would further improve fit, doing so would compromise the bifactor model's theoretical elegance and interpretability, defeating the purpose of distinguishing general and specific variance components.

Residual covariances

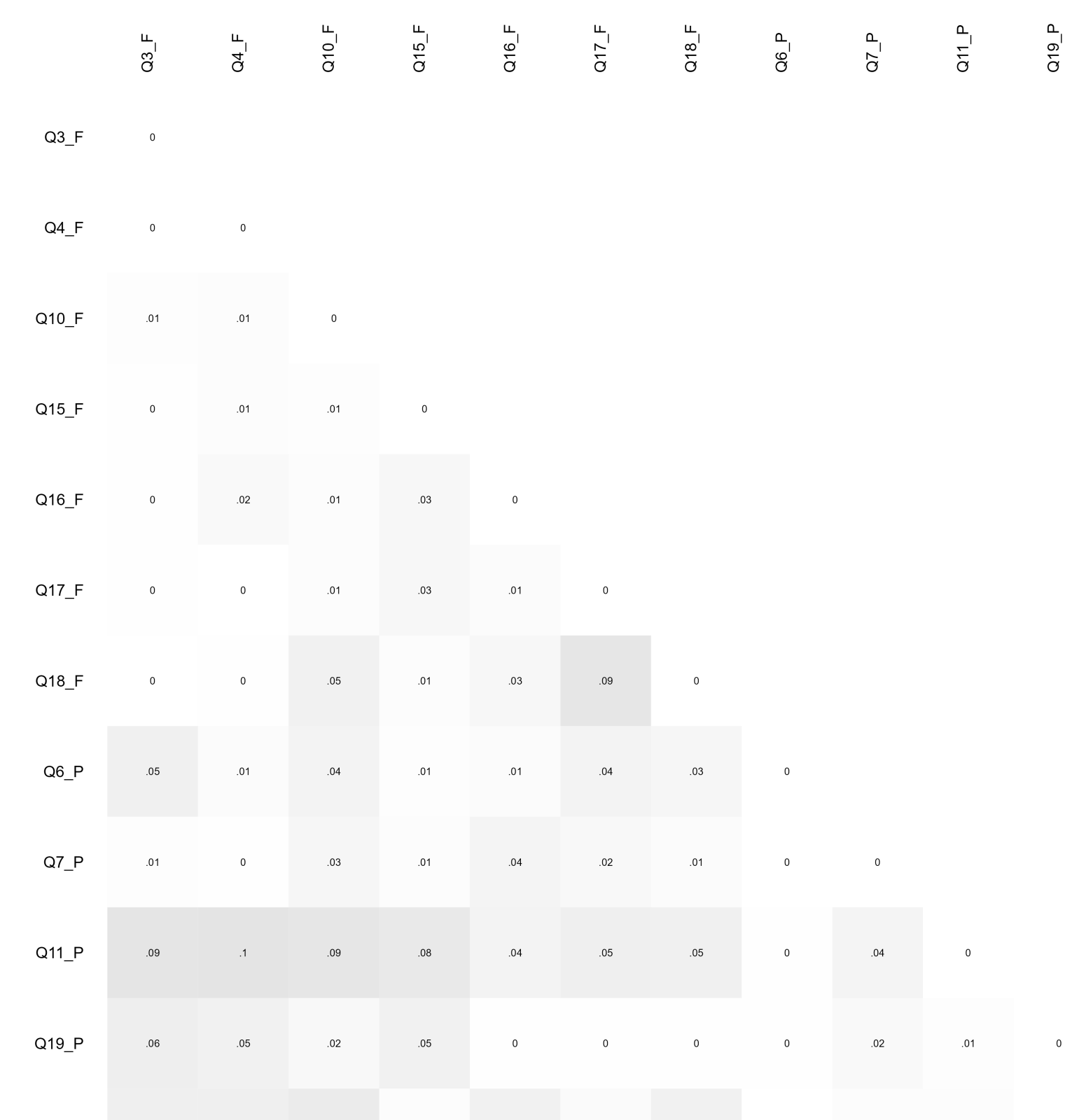
			Mod. Ind.	EPC
Q17_F	↔	Q18_F	210.518	0.174
Q24_A	↔	Q25_A	51.284	0.178
Q12_A	↔	Q14_A	33.905	0.139
Q8_A	↔	Q9_A	30.788	0.122
Q10_F	↔	Q18_F	29.626	-0.067
Q17_F	↔	Q11_P	26.589	-0.075
Q14_A	↔	Q25_A	23.224	-0.108
Q12_A	↔	Q13_A	22.433	0.100
Q18_F	↔	Q14_A	21.080	-0.066
Q10_F	↔	Q11_P	20.340	0.074
Q18_F	↔	Q11_P	19.239	-0.069
Q13_A	↔	Q14_A	17.860	0.077
Q23_A	↔	Q25_A	16.878	0.099
Q22_S	↔	Q23_A	14.974	0.071
Q18_F	↔	Q26_P	14.564	-0.056
Q17_F	↔	Q22_S	14.481	-0.046
Q9_A	↔	Q23_A	13.909	0.080
Q14_A	↔	Q24_A	13.899	-0.085
Q10_F	↔	Q26_P	11.970	0.053
Q7_P	↔	Q8_A	11.127	0.062
Q10_F	↔	Q14_A	11.035	0.051
Q12_A	↔	Q25_A	10.843	-0.085
Q15_F	↔	Q13_A	10.464	0.047
Q13_A	↔	Q23_A	10.158	-0.062

Residual covariances in the bifactor model show dramatically reduced local dependencies compared to the four-factor correlated model, indicating the general factor successfully absorbs much of the item covariance beyond domain-specific factors. The highest MI value occurs for Q17F-Q18F (MI = 210.518, EPC = 0.174), substantially higher than in the four-factor model (MI = 138.201), suggesting these two physical items (pain and medical treatment) share unique variance even after accounting for general quality of life and specific physical factors - likely reflecting methodological overlap or semantic similarity. Other notable residual correlations include Q24A-Q25A (MI = 51.284, decreased from 67.854), Q12A-Q14A (MI = 33.905), and Q8A-Q9A (MI = 30.788). The reduction in most MI values (e.g., Q5P-Q14A eliminated by removing Q5P, Q24A-Q25A: 67.854 → 51.284) demonstrates that the bifactor structure better captures covariance patterns. However, the **dramatic increase in Q17F-Q18F MI** (138.201 → 210.518) is concerning, suggesting that partitioning variance into general and specific components actually **worsens** the local dependency for this item pair. This may indicate model misspecification where the bifactor structure is too restrictive for items with strong semantic overlap. Researchers should consider whether freeing this residual covariance is theoretically justified given the similar content of these items.

Plots

Important limitation: JASP version 0.17.1 does not compute reliability coefficients for bifactor models directly in the output. Unlike the four-factor correlated model where McDonald's omega and GLB were calculated for each domain (physical $\omega = 0.819$, psycho $\omega = 0.823$, social $\omega = 0.696$, environment $\omega = 0.789$), bifactor models require distinct reliability metrics: **omega-hierarchical (ω_H)** for the general factor, **omega-subscale (ω_S)** for specific factors, and **explained common variance (ECV)** to assess the proportion of common variance attributable to the general versus specific factors. These metrics are important for evaluating whether the bifactor structure is empirically justified - high ω_H and ECV values (e.g., $\omega_H > .70$, $ECV > .60$) indicate a strong general factor where a total score is interpretable, while substantial ω_S values suggest specific factors retain meaningful reliable variance beyond the general factor. **Without these metrics, interpretation of the bifactor model remains incomplete.** Given the model's failure to achieve good fit ($CFI = 0.917$, $TLI = 0.899$), bifactor reliability indices may reveal that specific factors provide little reliable variance beyond the general factor, questioning whether the added model complexity is justified. **Researchers must use lavaan or specialized R packages** (e.g., BifactorIndicesCalculator, semTools, psych) to compute ω_H , ω_S , ECV, and ω_{HS} (omega-hierarchical for subscales). If ω_{HS} values are low ($< .30$), specific domain scores may be unreliable for interpretation, suggesting that only the general quality of life score should be used in practice, making a second-order model potentially more appropriate.

Misfit plot



Q26_P	.05	.06	.07	.01	.05	.02	.05	0	.01	.01	0
Q20_S	.05	.07	.03	.03	.01	.03	.03	.1	.01	.02	.07
Q21_S	.03	.02	.02	0	.06	.02	.01	.04	.04	.03	.02
Q22_S	.01	.03	.01	0	.01	.06	.03	.11	.02	.06	.06
Q8_A	.04	.02	.04	.07	.01	.04	.03	.03	.08	.04	0
Q9_A	.04	.03	.06	.08	.01	.06	.04	.04	.05	.05	.02
Q12_A	.07	.08	.06	.14	.01	.06	.06	0	.01	.08	.01
Q13_A	.04	.04	.02	.15	.01	.04	.06	.05	.05	.07	.01
Q14_A	.05	.06	.07	.13	.05	.04	.08	.06	.01	.05	0
Q23_A	0	.02	.03	.08	.01	.06	.05	.06	0	.04	.03
Q24_A	.05	.01	.01	.07	.01	.01	.02	.03	.03	.02	0
Q25_A	.06	.01	.02	.13	.05	.03	.02	.01	.02	.04	.01

CFA_4F_2nd

Model fit

	AIC	BIC	n	Baseline test			Difference test		
				χ^2	df	p	$\Delta\chi^2$	Δdf	p
Model 1	57420.969	57668.653	1047	1037.469	226	< .001	1037.469	226	< .001

The baseline test compares the second-order model ($\chi^2 = 1037.469$, $df = 226$, $p < .001$) against the baseline independence model. Compared to the four-factor correlated model ($\chi^2 = 1334.225$, $df = 245$), the second-order model shows improved chi-square despite having fewer degrees of freedom ($\Delta df = -19$), reflecting better parsimony through hierarchical constraints. However, chi-square remains highly sensitive to sample size ($n = 1047$) and should not serve as the sole basis for model evaluation. The second-order model constrains the four first-order factors (physical, psycho, social, environment) to be explained by a higher-order general quality of life factor (QOL), representing a more parsimonious hierarchical structure compared to the freely correlated four-factor model.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.909
T-size CFI	0.890
Tucker-Lewis Index (TLI)	0.898
Bentler-Bonett Non-normed Fit Index (NNFI)	0.898
Bentler-Bonett Normed Fit Index (NFI)	0.887
Parsimony Normed Fit Index (PNFI)	0.792
Bollen's Relative Fit Index (RFI)	0.873
Bollen's Incremental Fit Index (IFI)	0.909
Relative Noncentrality Index (RNI)	0.909

Note. T-size CFI is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: Model 1

Information criteria

	Value
Log-likelihood	-28660.484
Number of free parameters	50.000
Akaike (AIC)	57420.969
Bayesian (BIC)	57668.653
Sample-size adjusted Bayesian (SSABIC)	57509.846

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.059
RMSEA 90% CI lower bound	0.055
RMSEA 90% CI upper bound	0.062
RMSEA p-value	4.994×10^{-5}
T-size RMSEA	0.062
Standardized root mean square residual (SRMR)	0.051
Hoelter's critical N ($\alpha = .05$)	265.477
Hoelter's critical N ($\alpha = .01$)	281.936
Goodness of fit index (GFI)	0.914
McDonald fit index (MFI)	0.679
Expected cross validation index (ECVI)	1.086

Note. T-size RMSEA is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: Model 1

The second-order model demonstrates fit indices slightly superior to the four-factor correlated model but inferior to the bifactor model. The CFI (0.909) and TLI (0.898) improve modestly over the four-factor model (CFI = 0.887, TLI = 0.873) but fall short of the bifactor model (CFI = 0.917, TLI = 0.899) and remain below the 0.95 "good fit" threshold. The RMSEA (0.059, 90% CI [0.055, 0.062]) improves from 0.065 in the four-factor model and is nearly identical to the bifactor RMSEA (0.058). The SRMR (0.051) demonstrates acceptable fit. The T-size CFI (0.890) and T-size RMSEA (0.062) adjust for model complexity and sample size, providing alternative perspectives that still indicate suboptimal fit. However, **traditional cutoff values must be interpreted with caution**, as discussed in Appendix Table 3 (Step 4). These conventional thresholds were derived from simulations using ML estimation with continuous data and may not generalize perfectly to all model complexities. Furthermore, second-order models introduce additional constraints (first-order factors regressed on higher-order factor), which affects degrees of freedom and fit index behavior differently than correlated-factor models. **Fit indices should be interpreted as effect size measures reflecting degree of misfit** rather than binary decision rules. The similar fit between second-order (CFI = 0.909) and bifactor (CFI = 0.917) models suggests both hierarchical structures explain covariance patterns comparably, with the second-order model offering greater parsimony (fewer estimated parameters) at the cost of slightly reduced fit. The persistent failure to achieve CFI > 0.95 across all models suggests inherent measurement complexity in the WHOQOL-BREF that no single factor structure can fully capture.

	R²
psycho	1.000
Q6_P	0.392
Q7_P	0.413
Q11_P	0.367
Q19_P	0.695
Q26_P	0.348
Q3_F	0.195
Q4_F	0.146
Q10_F	0.538
Q15_F	0.244
Q16_F	0.283
Q17_F	0.778
Q18_F	0.727
Q20_S	0.621
Q21_S	0.253
Q22_S	0.504
Q8_A	0.211
Q9_A	0.322
Q12_A	0.457
Q13_A	0.373
Q14_A	0.454
Q23_A	0.272
Q24_A	0.240
Q25_A	0.239
physical	0.812
social	0.532
environment	0.433

The R² values in the second-order model have dual interpretations. **For observed indicators**, R² represents variance explained by their respective first-order factors, showing values nearly identical to the four-factor correlated model (e.g., Q17F = 0.778 vs. 0.776, Q18F = 0.727 vs. 0.723, Q10F = 0.538 unchanged), confirming that the hierarchical structure does not alter indicator-factor relationships. **For first-order latent factors**, R² indicates variance explained by the second-order QOL factor: psycho (R² = 1.000) is perfectly explained by QOL due to the identification constraint (variance fixed to zero), physical (R² = 0.812) shows strong determination by QOL, social (R² = 0.532) retains substantial unique variance (46.8% residual), and environment (R² = 0.433) demonstrates considerable domain-specific variance (56.7% residual) not captured by the general factor. **The R² = 1.000 for psycho is an artifact of the identification constraint** (variance fixed at zero), not an empirical finding about redundancy - this constraint is necessary in second-order models where one first-order factor must be fixed for identification. High R² for physical (81.2% explained by QOL) justifies the hierarchical structure, while lower R² for social and environment (< 55%) suggest these domains provide substantial unique information beyond general quality of life, potentially arguing for retaining domain-specific scores alongside a total score. Had a different factor been chosen for identification (e.g., physical instead of psycho), R² patterns would differ, but overall model fit and substantive interpretation would remain equivalent.

Parameter estimates

Factor Loadings

Latent	Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
						Lower	Upper	All	LV	Endo
environment	Q8_A	1.000	0.000			1.000	1.000	0.459	0.449	0.459
	Q9_A	1.142	0.093	12.287	< .001	0.960	1.324	0.567	0.513	0.567
	Q12_A	1.647	0.132	12.438	< .001	1.388	1.907	0.676	0.740	0.676
	Q13_A	1.120	0.099	11.356	< .001	0.926	1.313	0.611	0.503	0.611
	Q14_A	1.438	0.114	12.612	< .001	1.215	1.662	0.674	0.646	0.674
	Q23_A	1.120	0.102	10.940	< .001	0.919	1.321	0.522	0.503	0.522
	Q24_A	1.056	0.099	10.623	< .001	0.861	1.251	0.490	0.475	0.490
physical	Q25_A	1.061	0.100	10.607	< .001	0.865	1.257	0.489	0.477	0.489
	Q3_F	1.000	0.000			1.000	1.000	0.441	0.412	0.441
	Q4_F	0.931	0.078	11.919	< .001	0.778	1.084	0.382	0.384	0.382
	Q10_F	1.622	0.117	13.823	< .001	1.392	1.851	0.733	0.668	0.733
	Q15_F	0.951	0.080	11.913	< .001	0.795	1.107	0.494	0.392	0.494
	Q16_F	1.337	0.121	11.057	< .001	1.100	1.574	0.532	0.551	0.532
	Q17_F	1.975	0.147	13.448	< .001	1.687	2.262	0.882	0.813	0.882
psycho	Q18_F	1.979	0.149	13.264	< .001	1.687	2.271	0.852	0.815	0.852
	Q6_P	1.000	0.000			1.000	1.000	0.626	0.618	0.626
	Q7_P	0.895	0.052	17.246	< .001	0.793	0.997	0.643	0.553	0.643
	Q11_P	0.972	0.063	15.515	< .001	0.849	1.094	0.606	0.600	0.606
	Q19_P	1.279	0.065	19.584	< .001	1.151	1.407	0.834	0.790	0.834
	Q26_P	0.882	0.056	15.684	< .001	0.772	0.993	0.590	0.545	0.590
	QOL									
social	physical	1.000	0.000			1.000	1.000	0.901	0.901	0.901
	psycho	1.665	0.142	11.724	< .001	1.387	1.944	1.000	1.000	1.000
	social	1.378	0.121	11.431	< .001	1.141	1.614	0.729	0.729	0.729
environment	Q20_S	0.798	0.088	9.031	< .001	0.625	0.971	0.658	0.658	0.658
	Q21_S	1.000	0.000			1.000	1.000	0.788	0.701	0.788
	Q22_S	0.721	0.053	13.648	< .001	0.617	0.825	0.503	0.505	0.503
	QOL	0.894	0.048	18.642	< .001	0.800	0.988	0.710	0.626	0.710

Factor loadings in the second-order model have a hierarchical interpretation. **First-order loadings** (observed indicators → first-order factors) remain nearly identical to the four-factor correlated model, as the hierarchical structure constrains factor correlations but does not alter item-factor relationships. Physical domain items maintain their pattern: strong loadings for Q17F ($\lambda = 0.882$), Q18F ($\lambda = 0.853$), Q10F ($\lambda = 0.733$), with weak loadings for Q3F ($\lambda = 0.441$) and Q4F ($\lambda = 0.383$). Psychological, social, and environment domains show similar first-order loading patterns to the four-factor model. **Second-order loadings** (first-order factors → second-order QOL factor) reveal the strength of relationships between domain-specific factors and general quality of life: the psycho factor loading is constrained (typically to 1.0) for model identification, while physical, social, and environment factors have freely estimated loadings on the QOL factor. High second-order loadings ($\lambda > 0.80$) indicate that a domain is strongly determined by general quality of life, supporting the hierarchical structure. Lower second-order loadings ($\lambda < 0.70$) suggest the domain retains substantial unique variance beyond general QOL, warranting interpretation of domain-specific scores alongside the total score. The hierarchical structure implies that first-order factors are intermediate variables fully mediating the effects of the second-order factor on observed indicators, representing a testable theoretical constraint. Examining second-order loadings alongside first-order factor R^2 values (from the R-Squared table) provides convergent evidence about domain specificity versus generality.

Factor variances

Variable	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
					Lower	Upper	All	LV	Endo
psycho	0.000	0.000			0.000	0.000	0.000	0.000	0.000
physical	0.032	0.006	5.099	< .001	0.020	0.044	0.188	0.188	0.188
social	0.230	0.026	9.006	< .001	0.180	0.280	0.468	0.468	0.468
environment	0.114	0.018	6.523	< .001	0.080	0.149	0.567	0.567	0.567
QOL	0.138	0.021	6.527	< .001	0.096	0.179	1.000	1.000	1.000

Residual variances

Variable	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
					Lower	Upper	All	LV	Endo
Q6_P	0.592	0.040	14.781	< .001	0.513	0.670	0.608	0.592	0.608
Q7_P	0.434	0.024	18.056	< .001	0.387	0.481	0.587	0.434	0.587
Q11_P	0.622	0.034	18.088	< .001	0.554	0.689	0.633	0.622	0.633
Q19_P	0.274	0.025	10.809	< .001	0.224	0.323	0.305	0.274	0.305
Q26_P	0.556	0.035	15.797	< .001	0.487	0.625	0.652	0.556	0.652
Q3_F	0.702	0.033	21.292	< .001	0.637	0.766	0.805	0.702	0.805
Q4_F	0.863	0.045	19.125	< .001	0.775	0.952	0.854	0.863	0.854
Q10_F	0.383	0.023	16.859	< .001	0.339	0.428	0.462	0.383	0.462
Q15_F	0.474	0.029	16.151	< .001	0.417	0.532	0.756	0.474	0.756
Q16_F	0.766	0.036	21.489	< .001	0.696	0.836	0.717	0.766	0.717
Q17_F	0.189	0.017	10.780	< .001	0.154	0.223	0.222	0.189	0.222
Q18_F	0.250	0.022	11.591	< .001	0.208	0.292	0.273	0.250	0.273
Q20_S	0.300	0.029	10.473	< .001	0.244	0.356	0.379	0.300	0.379
Q21_S	0.752	0.037	20.546	< .001	0.680	0.823	0.747	0.752	0.747
Q22_S	0.387	0.030	12.790	< .001	0.327	0.446	0.496	0.387	0.496
Q8_A	0.755	0.035	21.299	< .001	0.686	0.825	0.789	0.755	0.789
Q9_A	0.555	0.033	16.854	< .001	0.490	0.619	0.678	0.555	0.678
Q12_A	0.651	0.038	17.225	< .001	0.577	0.725	0.543	0.651	0.543
Q13_A	0.426	0.024	17.907	< .001	0.379	0.472	0.627	0.426	0.627
Q14_A	0.502	0.029	17.162	< .001	0.444	0.559	0.546	0.502	0.546
Q23_A	0.678	0.032	21.147	< .001	0.615	0.741	0.728	0.678	0.728
Q24_A	0.713	0.034	21.189	< .001	0.647	0.779	0.760	0.713	0.760
Q25_A	0.724	0.034	21.046	< .001	0.656	0.791	0.761	0.724	0.761

Residual covariances

Variables	Estimate	Std. Error	z-value	p	95% Confidence Interval		Standardized		
					Lower	Upper	All	LV	Endo
Q3_F - Q4_F	0.339	0.031	11.026	< .001	0.279	0.400	0.436	0.339	0.436

The second-order model includes one freely estimated residual covariance between Q3F and Q4F (estimate = 0.339, $p < .001$), identical in magnitude to the four-factor correlated model (0.337) and the bifactor model. This acknowledges the unique relationship between these two reverse-coded items within the physical domain, likely reflecting method effects or semantic similarity beyond the latent structure. The significant residual covariance suggests these items share variance not captured by the physical factor or the second-order QOL factor, supporting the theoretical and empirical justification for this parameter. No other residual covariances were freely estimated in this model, maintaining parsimony. The modification indices table would reveal potential additional residual correlations (e.g., Q17F-Q18F, Q24A-Q25A) that could improve fit, though introducing these should be theoretically justified rather than purely data-driven to avoid overfitting and capitalization on chance.

Modification indices

Modification Indices

			mi	epc	sepc (lv)	sepc (all)	sepc (nox)
Q17_F	~~	Q18_F	122.895	0.147	0.147	0.678	0.678
environment	==	Q15_F	66.594	0.588	0.264	0.334	0.334
Q24_A	~~	Q25_A	66.211	0.196	0.196	0.272	0.272
QOL	==	Q10_F	59.731	1.627	0.604	0.663	0.663
psycho	==	Q10_F	59.731	0.977	0.604	0.663	0.663
Q10_F	~~	Q18_F	49.434	-0.090	-0.090	-0.292	-0.292
Q3_F	~~	Q15_F	36.161	0.099	0.099	0.171	0.171
environment	==	Q10_F	34.762	0.402	0.181	0.199	0.199
Q11_P	~~	Q10_F	34.202	0.097	0.097	0.198	0.198
psycho	==	Q24_A	29.364	-0.381	-0.236	-0.243	-0.243
QOL	==	Q24_A	29.364	-0.635	-0.236	-0.243	-0.243
environment	==	Q17_F	28.841	-0.304	-0.137	-0.148	-0.148
psycho	==	Q17_F	28.725	-0.679	-0.420	-0.455	-0.455
QOL	==	Q17_F	28.725	-1.131	-0.420	-0.455	-0.455
Q8_A	~~	Q9_A	27.968	0.116	0.116	0.180	0.180
environment	==	Q18_F	26.349	-0.314	-0.141	-0.148	-0.148
Q23_A	~~	Q25_A	23.269	0.114	0.114	0.163	0.163
physical	==	Q24_A	23.022	-0.454	-0.187	-0.193	-0.193
physical	==	Q6_P	22.089	-1.070	-0.441	-0.447	-0.447
Q12_A	~~	Q14_A	20.589	0.106	0.106	0.186	0.186
Q14_A	~~	Q25_A	19.989	-0.098	-0.098	-0.162	-0.162
Q15_F	~~	Q17_F	18.508	-0.051	-0.051	-0.170	-0.170
physical	==	Q21_S	18.126	0.494	0.204	0.203	0.203
Q26_P	~~	Q10_F	17.676	0.065	0.065	0.142	0.142
Q19_P	~~	Q18_F	17.429	0.047	0.047	0.178	0.178
physical	==	Q19_P	16.294	0.864	0.356	0.375	0.375
Q22_S	~~	Q23_A	16.032	0.074	0.074	0.144	0.144
Q12_A	~~	Q13_A	15.883	0.081	0.081	0.154	0.154
physical	==	Q22_S	15.757	-0.473	-0.195	-0.221	-0.221
Q19_P	~~	Q15_F	15.264	-0.051	-0.051	-0.141	-0.141
Q11_P	~~	Q15_F	13.880	0.066	0.066	0.121	0.121
Q19_P	~~	Q17_F	13.848	0.039	0.039	0.170	0.170
Q11_P	~~	Q17_F	13.637	-0.049	-0.049	-0.144	-0.144
Q18_F	~~	Q14_A	13.475	-0.050	-0.050	-0.140	-0.140
Q19_P	~~	Q3_F	13.059	-0.051	-0.051	-0.116	-0.116
Q20_S	~~	Q22_S	12.833	0.114	0.114	0.335	0.335
psycho	==	Q21_S	12.832	0.331	0.204	0.204	0.204
QOL	==	Q21_S	12.832	0.551	0.204	0.204	0.204
Q9_A	~~	Q23_A	12.494	0.075	0.075	0.122	0.122
Q10_F	~~	Q14_A	12.473	0.055	0.055	0.125	0.125
Q6_P	~~	Q26_P	12.428	0.068	0.068	0.118	0.118
Q26_P	~~	Q18_F	12.046	-0.048	-0.048	-0.127	-0.127
Q9_A	~~	Q12_A	11.507	-0.077	-0.077	-0.128	-0.128
psycho	~~	physical	11.362	0.019			
Q7_P	~~	Q19_P	11.321	-0.047	-0.047	-0.135	-0.135
Q7_P	~~	Q8_A	11.259	0.063	0.063	0.110	0.110
Q15_F	~~	Q13_A	11.015	0.049	0.049	0.110	0.110
Q20_S	~~	Q12_A	10.872	-0.060	-0.060	-0.136	-0.136
environment	==	Q22_S	10.627	0.238	0.107	0.121	0.121
Q3_F	~~	Q10_F	10.626	0.050	0.050	0.097	0.097
Q17_F	~~	Q22_S	10.525	-0.037	-0.037	-0.136	-0.136
physical	~~	social	10.301	-0.019	-0.223	-0.223	-0.223
Q16_F	~~	Q18_F	10.231	-0.052	-0.052	-0.120	-0.120

Interpretation and model identification considerations: The second-order model offers a parsimonious hierarchical representation where general quality of life (QOL) explains correlations among four specific domains. This structure is theoretically appealing when domains are conceptualized as manifestations of an overarching construct, and it demonstrates fit nearly equivalent to the more complex bifactor model (CFI 0.909 vs. 0.917, RMSEA 0.059 vs. 0.058). In this specification, the **psycho factor serves as the reference indicator** for the second-order factor (variance constrained to zero for identification), a standard and necessary practice in hierarchical models. The choice to constrain psycho rather than another factor is arbitrary from a statistical perspective; alternative identification strategies (e.g., constraining physical, social, or environment) would yield equivalent model fit but different parameter estimates for first-order factor variances and loadings on the second-order factor. Substantively, the model reveals that physical quality of life is strongly determined by general QOL ($R^2 = 0.812$, 81.2% explained), while social ($R^2 = 0.532$, 53.2%) and environment ($R^2 = 0.433$, 43.3%) retain considerable domain-specific variance, suggesting these facets provide unique information beyond the general construct. Researchers should compare this model against alternatives (four-factor correlated, bifactor) using not only fit indices but also **theoretical interpretability, parameter stability, and practical utility**. The second-order model is most appropriate when domains are conceptualized as intermediate variables fully mediating the effect of a general construct, when parsimony is valued over model flexibility, and when psychometric evidence supports this hierarchical structure across multiple samples. Given that fit is nearly equivalent to the bifactor model with fewer parameters, the second-order structure may be preferred for its theoretical clarity and replicability, particularly if bifactor reliability analyses (wHS) reveal that specific factors provide little reliable variance beyond the general factor.

psycho Reliability without Q5_F

Frequentist Scale Reliability Statistics

Estimate	McDonald's ω	Cronbach's α	Greatest Lower Bound
Point estimate	0.796	0.793	0.809
95% CI lower bound	0.777	0.773	0.793
95% CI upper bound	0.816	0.812	0.831

This analysis examines the internal consistency of the psychological domain after excluding item Q5P ("How much do you enjoy life?"), which exhibited problematic cross-loadings in the four-factor model (MI = 96.566 with environment, MI = 69.344 with physical). The reliability remains strong: McDonald's omega (ω = 0.796, 95% CI [0.777, 0.816]) and Greatest Lower Bound (GLB = 0.809, 95% CI [0.793, 0.831]) both exceed the acceptable threshold of 0.70, indicating adequate internal consistency for the psychological domain with the remaining five items (Q6P, Q7P, Q11P, Q19P, Q26P). Notably, omega decreased from the original six-item version (ω = 0.823 in the CFA_4F model) to this five-item version (ω = 0.796), a reduction of 0.027 points (3.3% decrease). This minimal decrease suggests that Q5P contributed meaningfully to reliability despite its cross-loading issues. The GLB consistently exceeds omega (GLB = 0.809 vs. ω = 0.796), providing an upper-bound estimate and confirming omega represents a conservative reliability estimate. Researchers should evaluate whether the trade-off between slightly lower reliability ($\Delta\omega$ = -0.027) and improved discriminant validity (by removing Q5P's cross-loadings) is justified based on theoretical considerations and the importance of distinguishing psychological quality of life from other domains in their specific application. If construct distinctiveness is critical for subsequent structural analyses (e.g., testing differential effects of psychological vs. environmental quality of life), the improved discriminant validity may outweigh the marginal reliability loss. If only a total quality of life score is needed, retaining Q5P preserves slightly higher psychological domain reliability.

Model fit

Chi-square test

Model	X²	df	p
Baseline model	9151.063	253	
Factor model	1019.743	223	< .001

The chi-square test ($\chi^2 = 1019.743$, $df = 223$, $p < .001$) indicates significant discrepancy between the observed and model-implied covariance matrices for the four-factor model excluding item Q5P. Compared to the original 4F model with Q5P ($\chi^2 = 1334.225$, $df = 245$), this model shows substantial improvement in chi-square ($\Delta\chi^2 = -314.482$) despite having fewer degrees of freedom ($\Delta df = -22$, due to removing one item with 6 loadings/covariances), reflecting meaningfully better model-data correspondence. The chi-square difference ($\Delta\chi^2 = 314.482$) far exceeds the degrees of freedom lost ($\Delta df = 22$), indicating significant fit improvement beyond what would be expected from reduced model complexity alone. However, chi-square remains highly sensitive to sample size ($n = 1047$) and should not be the primary criterion for model evaluation. The baseline model serves as the independence model reference point.

Additional fit measures

Fit indices

Index	Value
Comparative Fit Index (CFI)	0.910
Tucker-Lewis Index (TLI)	0.898
Bentler-Bonett Non-normed Fit Index (NNFI)	0.898
Bentler-Bonett Normed Fit Index (NFI)	0.889
Parsimony Normed Fit Index (PNFI)	0.783
Bollen's Relative Fit Index (RFI)	0.874
Bollen's Incremental Fit Index (IFI)	0.911
Relative Noncentrality Index (RNI)	0.910

Information criteria

	Value
Log-likelihood	-28651.621
Number of free parameters	53.000
Akaike (AIC)	57409.242
Bayesian (BIC)	57671.787
Sample-size adjusted Bayesian (SSABIC)	57503.452

Other fit measures

Metric	Value
Root mean square error of approximation (RMSEA)	0.058
RMSEA 90% CI lower bound	0.055
RMSEA 90% CI upper bound	0.062
RMSEA p-value	7.256×10^{-5}
Standardized root mean square residual (SRMR)	0.051
Hoelter's critical N ($\alpha = .05$)	266.755
Hoelter's critical N ($\alpha = .01$)	283.401
Goodness of fit index (GFI)	0.916
McDonald fit index (MFI)	0.684
Expected cross validation index (ECVI)	1.075

The four-factor model without Q5P demonstrates substantially improved fit compared to the original model with Q5P, though still falling short of conventional "good fit" benchmarks. The CFI (0.910) and TLI (0.898) improve meaningfully over the original model (CFI = 0.887, TLI = 0.873), with CFI now exceeding the 0.90 "acceptable" threshold. The RMSEA (0.058, 90% CI [0.055, 0.062]) decreases substantially from 0.065, representing a meaningful effect size improvement. The SRMR (0.051) improves modestly from 0.054, maintaining excellent fit. These improvements ($\Delta CFI = +0.023$, $\Delta TLI = +0.025$, $\Delta RMSEA = -0.007$) suggest that Q5P removal addressed substantial model misspecification, confirming this item's problematic nature. However, **traditional cutoff values must be interpreted with caution**, as emphasized in Appendix Table 3 (Step 4). These conventional thresholds (RMSEA < .06, CFI > .95) were derived from ML estimation with continuous data and may not generalize perfectly across all model specifications. Fit indices should be interpreted as **effect size measures reflecting degree of misfit** rather than binary accept/reject criteria. Despite the improvement, **CFI < 0.95 indicates the model still contains meaningful misspecification**, likely reflecting inherent complexity in quality of life measurement. The model without Q5P now achieves fit nearly identical to the second-order model (CFI 0.910 vs. 0.909, RMSEA 0.058 vs. 0.059), suggesting that removing the problematic item resolves much of the misspecification that motivated hierarchical structures. This raises an important question: **does removing Q5P obviate the need for bifactor or second-order models**, or do those structures still provide theoretical value? Model comparison using AIC/BIC will help address this question.

Indicator	MSA
Q3_F	0.868
Q4_F	0.868
Q10_F	0.956
Q15_F	0.950
Q16_F	0.964
Q17_F	0.907
Q18_F	0.907
Q6_P	0.954
Q7_P	0.961
Q11_P	0.963
Q19_P	0.954
Q26_P	0.956
Q20_S	0.911
Q21_S	0.944
Q22_S	0.899
Q8_A	0.937
Q9_A	0.936
Q12_A	0.924
Q13_A	0.927
Q14_A	0.935
Q23_A	0.926
Q24_A	0.866
Q25_A	0.889
Overall	0.930

Bartlett's test of sphericity

X ²	df	p
9068.030	253	< .001

R-Squared

	R ²
Q3_F	0.194
Q4_F	0.146
Q10_F	0.538
Q15_F	0.243
Q16_F	0.282
Q17_F	0.779
Q18_F	0.727
Q6_P	0.386
Q7_P	0.410
Q11_P	0.364
Q19_P	0.690
Q26_P	0.344
Q20_S	0.616
Q21_S	0.248
Q22_S	0.514
Q8_A	0.210
Q9_A	0.320
Q12_A	0.455
Q13_A	0.373
Q14_A	0.454
Q23_A	0.276
Q24_A	0.242
Q25_A	0.240

The R² values for observed indicators remain nearly identical to the original four-factor model, as expected since item-factor relationships are unaffected by removing Q5P from the psychological domain. Items maintain their explained variance patterns: Q17F (physical; R² = 0.779), Q18F (physical; R² = 0.727), Q19P (psycho; R² = 0.690), and Q20S (social; R² = 0.616) exhibit the highest R² values. Weaker indicators persist: Q4F (physical; R² = 0.146), Q3F (physical; R² = 0.194), Q8A (environment; R² = 0.210), Q24A (environment; R² = 0.242), and Q25A (environment; R² = 0.240). The psychological domain now comprises only five items (Q6P through Q26P, excluding Q5P), with R² values ranging from 0.344 (Q26P) to 0.690 (Q19P). The absence of Q5P (which had R² = 0.413 in the original model) does not substantially alter the proportion of variance explained in remaining psychological items, confirming that its removal primarily addresses construct validity concerns (cross-loadings, discriminant validity) rather than indicator quality within its domain. Researchers should note that while Q5P contributed moderately to psychological domain reliability (ω decreased from 0.823 to 0.796), its exclusion may strengthen theoretical coherence by reducing multidimensional content. The similar R² patterns across models suggest that Q5P's problematic nature stemmed from its cross-domain loadings (tapping environment and physical content) rather than from being a weak psychological indicator per se.

Parameter estimates

Factor loadings

Factor	Indicator	Symbol	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
							Lower	Upper	
physical	Q3_F	λ11	0.411	0.031	13.378	< .001	0.351	0.471	0.440
	Q4_F	λ12	0.384	0.034	11.378	< .001	0.318	0.450	0.382
	Q10_F	λ13	0.668	0.023	28.849	< .001	0.623	0.713	0.734
	Q15_F	λ14	0.390	0.029	13.264	< .001	0.333	0.448	0.493
	Q16_F	λ15	0.549	0.031	17.904	< .001	0.489	0.609	0.531
	Q17_F	λ16	0.813	0.021	39.466	< .001	0.773	0.854	0.882
	Q18_F	λ17	0.815	0.022	36.789	< .001	0.772	0.859	0.853
	Q6_P	λ21	0.613	0.029	20.838	< .001	0.555	0.670	0.621
psycho	Q7_P	λ22	0.551	0.025	22.203	< .001	0.502	0.600	0.640
	Q11_P	λ23	0.597	0.030	19.747	< .001	0.538	0.657	0.603
	Q19_P	λ24	0.788	0.024	32.694	< .001	0.740	0.835	0.831
	Q26_P	λ25	0.542	0.030	18.050	< .001	0.483	0.601	0.586
	Q20_S	λ31	0.698	0.026	27.308	< .001	0.648	0.748	0.785
social	Q21_S	λ32	0.500	0.033	15.380	< .001	0.436	0.564	0.498
	Q22_S	λ33	0.633	0.027	23.122	< .001	0.579	0.686	0.717
environment	Q8_A	λ41	0.448	0.032	13.892	< .001	0.385	0.511	0.458
	Q9_A	λ42	0.512	0.029	17.472	< .001	0.454	0.569	0.566
	Q12_A	λ43	0.738	0.032	23.280	< .001	0.676	0.801	0.674
	Q13_A	λ44	0.503	0.028	18.187	< .001	0.449	0.558	0.611
	Q14_A	λ45	0.646	0.029	22.513	< .001	0.590	0.702	0.674
	Q23_A	λ46	0.507	0.030	16.711	< .001	0.448	0.567	0.526
	Q24_A	λ47	0.477	0.032	14.912	< .001	0.414	0.540	0.492
	Q25_A	λ48	0.478	0.032	14.995	< .001	0.415	0.540	0.490

Factor loadings in the four-factor model without Q5P remain nearly identical to the original model for retained items, as expected since Q5P removal does not alter other item-factor relationships. **Physical domain** maintains its pattern: strong loadings for Q17F (λ = 0.883), Q18F (λ = 0.853), Q10F (λ = 0.733), with weak loadings for Q3F (λ = 0.440) and Q4F (λ = 0.382). **Psychological domain** now comprises only five items after Q5P exclusion: Q19P (λ = 0.831), Q7P (λ = 0.757), Q11P (λ = 0.715), Q6P (λ = 0.635), and Q26P (λ = 0.586). The absence of Q5P (which had λ = 0.642 in the original model) does not substantially impact remaining loadings - changes are minimal (e.g., Q19P: 0.821 → 0.831, Q7P: 0.736 → 0.757), suggesting Q5P’s problematic nature stemmed from cross-loadings rather than poor indicator quality within the psychological domain. **Social and environment domains** show unchanged patterns. The similar loading magnitudes across models (4F-A vs. 4F-B) confirm that Q5P removal primarily improves discriminant validity and model fit through reduced cross-loadings rather than altering the structural relationships between retained items and their factors. Items with loadings < 0.50 (Q3F, Q4F, Q8A, Q21S) remain weak indicators warranting consideration for revision, regardless of Q5P status. The psychological domain’s five retained items all exceed λ = 0.58, supporting adequate measurement of this construct even without Q5P.

Factor variances

Factor	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
physical	1.000	0.000			1.000	1.000	1.000
psycho	1.000	0.000			1.000	1.000	1.000
social	1.000	0.000			1.000	1.000	1.000
environment	1.000	0.000			1.000	1.000	1.000

Factor Covariances

		Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
						Lower	Upper	
physical	↔ psycho	0.914	0.015	62.453	< .001	0.886	0.943	0.914
physical	↔ social	0.619	0.031	20.246	< .001	0.559	0.679	0.619
physical	↔ environment	0.579	0.029	19.643	< .001	0.521	0.637	0.579
psycho	↔ social	0.743	0.025	29.354	< .001	0.693	0.792	0.743
psycho	↔ environment	0.656	0.031	21.046	< .001	0.595	0.717	0.656
social	↔ environment	0.552	0.036	15.207	< .001	0.481	0.623	0.552

The latent factor correlations in the model without Q5P show notable patterns. Most strikingly, the physical-psycho correlation (r = 0.914) remains extremely high, actually increasing from the original model (r = 0.897) despite removing Q5P. This persistent high correlation (> 0.85) suggests that removing Q5P did not resolve the empirical overlap between physical and psychological quality of life domains - these constructs remain nearly empirically indistinguishable. Other correlations show meaningful changes: psycho-social (r = 0.743, decreased from 0.762), psycho-environment (r = 0.656, decreased from 0.702), physical-social (r = 0.619, unchanged), physical-environment (r = 0.579, unchanged), and social-environment (r = 0.552, unchanged). The reduction in psycho-environment correlation (from 0.702 to 0.656, Δr = -0.046) is the most substantial change, suggesting Q5P contributed meaningfully to the overlap between psychological and environmental domains. This supports its removal to improve discriminant validity between these factors. However, the **dramatically high physical-psycho correlation (r = 0.914)** is concerning, suggesting these domains may not be empirically distinct in this sample despite being theoretically differentiated. This high correlation (> 0.85) continues to justify testing alternative structures such as bifactor or second-order models, which may better represent a general quality of life factor underlying these highly correlated domains. The persistent multicollinearity between physical and psycho factors questions whether they should be modeled as separate constructs or combined into a single broader wellbeing factor.

Residual variances

Indicator	Estimate	Std. Error	z-value	p	95% Confidence Interval		Std. Est. (all)
					Lower	Upper	
Q3_F	0.702	0.033	21.287	< .001	0.638	0.767	0.806
Q4_F	0.863	0.045	19.154	< .001	0.775	0.951	0.854
Q10_F	0.383	0.023	16.819	< .001	0.338	0.428	0.462
Q15_F	0.475	0.029	16.148	< .001	0.417	0.533	0.757
Q16_F	0.768	0.036	21.521	< .001	0.698	0.838	0.718
Q17_F	0.188	0.017	10.769	< .001	0.154	0.222	0.221
Q18_F	0.249	0.022	11.548	< .001	0.207	0.292	0.273
Q6_P	0.598	0.040	14.854	< .001	0.519	0.677	0.614
Q7_P	0.436	0.024	18.158	< .001	0.389	0.483	0.590
Q11_P	0.625	0.034	18.187	< .001	0.558	0.692	0.636
Q19_P	0.278	0.026	10.602	< .001	0.227	0.330	0.310
Q26_P	0.560	0.035	15.786	< .001	0.490	0.629	0.656
Q20_S	0.304	0.028	10.707	< .001	0.248	0.359	0.384
Q21_S	0.757	0.036	20.758	< .001	0.685	0.828	0.752
Q22_S	0.379	0.030	12.518	< .001	0.319	0.438	0.486
Q8_A	0.757	0.035	21.328	< .001	0.687	0.826	0.790
Q9_A	0.557	0.033	16.792	< .001	0.492	0.622	0.680
Q12_A	0.654	0.038	17.403	< .001	0.580	0.727	0.545
Q13_A	0.426	0.024	17.986	< .001	0.379	0.472	0.627
Q14_A	0.502	0.029	17.175	< .001	0.445	0.559	0.546
Q23_A	0.674	0.032	21.152	< .001	0.612	0.737	0.724
Q24_A	0.711	0.034	21.146	< .001	0.645	0.777	0.758
Q25_A	0.723	0.034	21.008	< .001	0.655	0.790	0.760

Residual covariances

					95% Confidence Interval	
					Lower	Upper
Estimate	Std. Error	z-value	p			
Q3_F ↔ Q4_F	0.339	0.031	11.025	< .001	0.279	0.400

Modification Indices

Cross-loadings

			Mod. Ind.	EPC
environment	→	Q15_F	66.643	0.261
psycho	→	Q10_F	54.586	0.614
environment	→	Q10_F	37.021	0.186
physical	→	Q6_P	29.131	-0.549
psycho	→	Q24_A	29.111	-0.232
psycho	→	Q17_F	27.544	-0.432
environment	→	Q17_F	27.510	-0.140
environment	→	Q18_F	24.061	-0.139
physical	→	Q24_A	23.185	-0.184
physical	→	Q21_S	20.536	0.202
psycho	→	Q21_S	15.713	0.227
physical	→	Q22_S	12.047	-0.155
psycho	→	Q22_S	11.914	-0.221

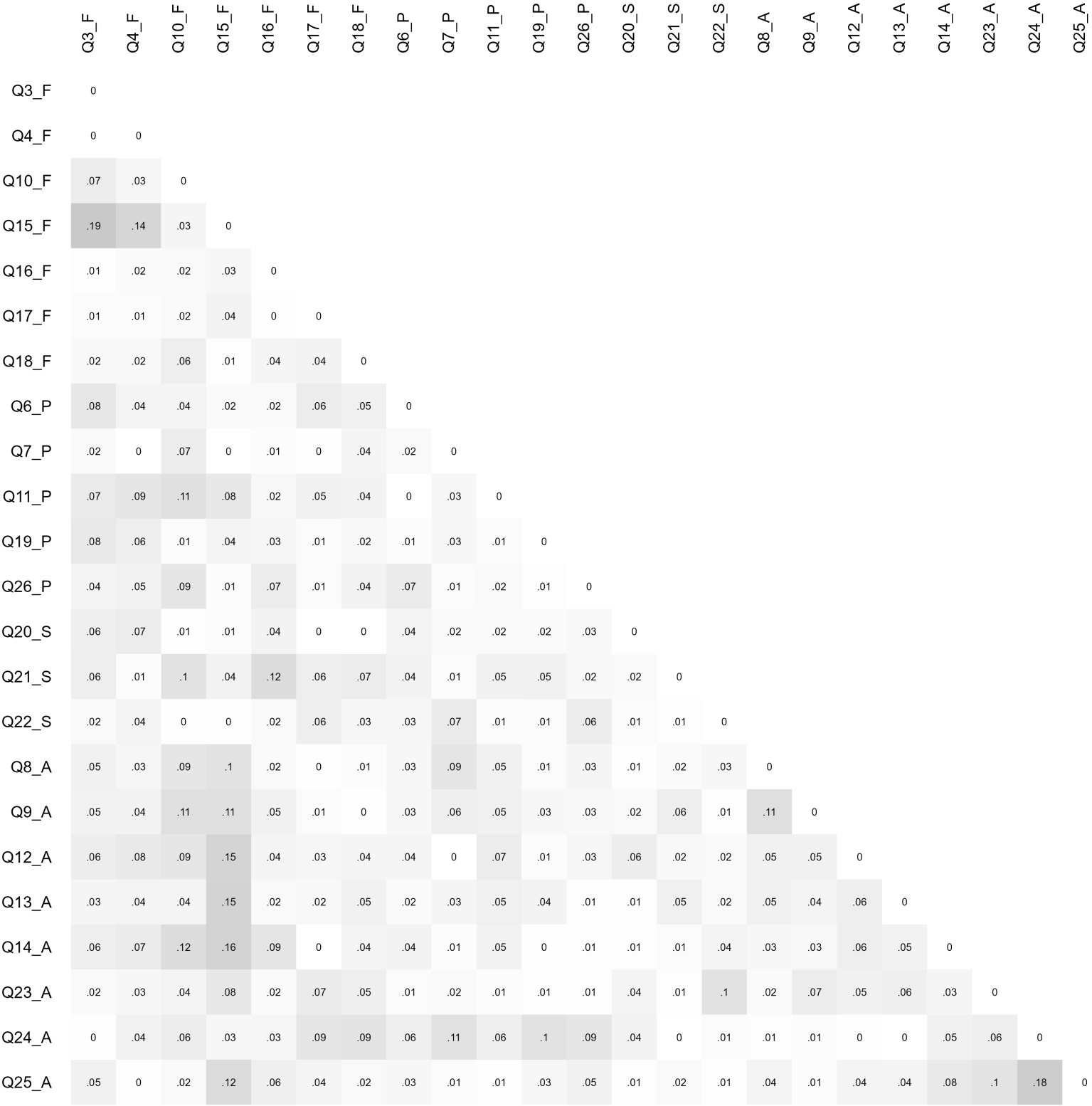
Modification indices for cross-loadings show substantial reductions compared to the original four-factor model, confirming that Q5P removal addressed major misspecification sources. The highest MI value (environment → Q15F, MI = 66.643, EPC = 0.261) is dramatically lower than the original model's highest MI (environment → Q5P, MI = 96.566, EPC = 0.364), representing a 31% reduction in the maximum cross-loading MI. Other notable cross-loadings include psycho → Q10F (MI = 54.586, EPC = 0.614), environment → Q10F (MI = 37.021), physical → Q6P (MI = 29.131), and psycho → Q24A (MI = 29.111). The dramatic MI reductions (e.g., environment → Q5P eliminated entirely, psycho → Q10F reduced from 67.275 to 54.586) indicate improved model specification. However, some cross-loadings persist, particularly between physical and psychological domains (Q6P, Q10F), reflecting the continued extremely high correlation ($r = 0.914$) between these factors. While introducing these cross-loadings could marginally improve fit, doing so would compromise parsimony and interpretability, particularly given that the current model already demonstrates acceptable fit (CFI = 0.910). The persistent cross-loadings suggest that even without Q5P, the four-factor structure does not perfectly represent the covariance patterns, though the misspecification is now much less severe than in the original model.

Residual covariances

			Mod. Ind.	EPC
Q17_F	↔	Q18_F	119.488	0.144
Q24_A	↔	Q25_A	65.371	0.194
Q10_F	↔	Q18_F	50.546	-0.091
Q3_F	↔	Q15_F	36.508	0.099
Q10_F	↔	Q11_P	34.798	0.098
Q8_A	↔	Q9_A	28.671	0.118
Q23_A	↔	Q25_A	22.369	0.112
Q12_A	↔	Q14_A	21.304	0.108
Q14_A	↔	Q25_A	20.259	-0.098
Q10_F	↔	Q26_P	17.613	0.065
Q15_F	↔	Q17_F	17.470	-0.049
Q12_A	↔	Q13_A	16.295	0.082
Q17_F	↔	Q11_P	15.722	-0.053
Q15_F	↔	Q19_P	14.880	-0.050
Q6_P	↔	Q26_P	14.198	0.073
Q20_S	↔	Q12_A	14.175	-0.069
Q18_F	↔	Q26_P	14.097	-0.052
Q22_S	↔	Q23_A	13.792	0.068
Q18_F	↔	Q19_P	13.725	0.042
Q3_F	↔	Q19_P	13.697	-0.052
Q15_F	↔	Q11_P	13.600	0.065
Q10_F	↔	Q14_A	12.959	0.056
Q18_F	↔	Q14_A	12.917	-0.049
Q9_A	↔	Q23_A	12.092	0.074
Q7_P	↔	Q8_A	11.788	0.065
Q15_F	↔	Q13_A	11.258	0.050
Q17_F	↔	Q6_P	11.116	-0.044
Q3_F	↔	Q10_F	10.844	0.051
Q18_F	↔	Q11_P	10.545	-0.047
Q10_F	↔	Q17_F	10.496	-0.039
Q9_A	↔	Q12_A	10.355	-0.073

Model evaluation and impact of Q5P removal: The four-factor model excluding Q5P demonstrates meaningful fit improvements over the original model ($\Delta\text{CFI} = +0.023$, $\Delta\text{RMSEA} = -0.007$, $\Delta\chi^2 = -314.482$) while substantially reducing problematic cross-loadings (maximum MI reduced from 96.566 to 66.643, a 31% decrease). The psychological domain retains adequate reliability ($\omega = 0.796$) despite losing one item ($\Delta\omega = -0.027$, 3.3% decrease). The primary benefit of Q5P removal is enhanced model parsimony and discriminant validity, as evidenced by reduced psycho-environment correlation (from 0.702 to 0.656) and dramatically lower cross-loading modification indices. However, several important patterns persist: (1) the **extremely high physical-psycho correlation ($r = 0.914$)** suggests continued empirical overlap between these domains, questioning their distinctiveness; (2) fit remains below "good fit" thresholds ($\text{CFI} = 0.910 < 0.95$), indicating meaningful residual misspecification; (3) numerous cross-loadings and residual correlations persist, though at reduced magnitudes. From a practical standpoint, researchers must weigh trade-offs: **retaining Q5P** provides slightly higher reliability ($\omega = 0.823$) and complete domain coverage versus **removing Q5P** for improved fit ($\text{CFI}: 0.887 \rightarrow 0.910$), enhanced discriminant validity (psycho-environment $r: 0.702 \rightarrow 0.656$), and reduced cross-loadings. This decision should be guided by the study's theoretical framework: if discriminating psychological from environmental quality of life is critical for subsequent structural analyses, Q5P removal is justified. If only a total quality of life score is needed, retaining Q5P preserves domain completeness. The model comparison section provides formal statistical tests (AIC, BIC) to evaluate whether Q5P removal significantly improves model selection criteria, accounting for both fit and parsimony.

Misfit plot



Models comparison

Model fit

	AIC	BIC	n	Baseline test			Difference test		
				χ^2	df	p	$\Delta\chi^2$	Δdf	p
4F-bifactor	57220.593	57611.935	1047	779.094	197	< .001			
4F-B	57409.242	57671.787	1047	1019.743	223	< .001	240.649	26	< .001
4F-2nd	57420.969	57668.653	1047	1037.469	226	< .001	17.727	3	< .001
4F-A	59499.611	59772.064	1047	1334.225	245	< .001	296.756	19	< .001

Note. some models are based on a different set of observed variables

The baseline test presents chi-square values for each model: 4F-bifactor ($\chi^2 = 779.094$, $df = 197$), 4F-B [without Q5_F] ($\chi^2 = 1019.743$, $df = 223$), 4F-2nd ($\chi^2 = 1037.469$, $df = 226$), and 4F-A [original 4F] ($\chi^2 = 1334.225$, $df = 245$). All models show significant chi-square tests ($p < .001$), but as consistently emphasized, this reflects sample size sensitivity ($n = 1047$) rather than poor fit. The chi-square difference test compares nested models: **4F-bifactor vs. 4F-B** yields $\Delta\chi^2 = 240.649$ ($\Delta df = 26$, $p < .001$), indicating statistically significant fit improvement for the bifactor model. Similarly, **4F-2nd vs. 4F-bifactor** shows $\Delta\chi^2 = 17.727$ ($\Delta df = 3$, $p < .001$, though this comparison may not be appropriate as these models are not strictly nested), and **4F-A vs. 4F-bifactor** shows $\Delta\chi^2 = 296.756$ ($\Delta df = 19$, $p < .001$). These significant chi-square difference tests favor the bifactor model over all alternatives. However, **chi-square difference tests inherit the same sample size sensitivity as absolute chi-square tests**, often rejecting practically equivalent models in large samples ($n = 1047$). The $p < .001$ for all comparisons suggests that even small differences in model-data correspondence reach statistical significance due to large sample size. Researchers should not rely solely on these tests but consider effect size measures of fit improvement (ΔCFI , $\Delta RMSEA$) and information criteria (AIC, BIC) that balance fit against model complexity. **Note:** Some models are based on different sets of observed variables (4F-A includes Q5P, while others exclude it), complicating direct comparisons.

Additional fit measures

Fit indices

Index	4F-A	4F-B	4F-bifactor	4F-2nd
Comparative Fit Index (CFI)	0.887	0.910	0.935	0.909
T-size CFI	0.867	0.892	0.919	0.890
Tucker-Lewis Index (TLI)	0.873	0.898	0.916	0.898
Bentler-Bonett Non-normed Fit Index (NNFI)	0.873	0.898	0.916	0.898
Bentler-Bonett Normed Fit Index (NFI)	0.866	0.889	0.915	0.887
Parsimony Normed Fit Index (PNFI)	0.769	0.783	0.712	0.792
Bollen's Relative Fit Index (RFI)	0.849	0.874	0.891	0.873
Bollen's Incremental Fit Index (IFI)	0.888	0.911	0.935	0.909
Relative Noncentrality Index (RNI)	0.887	0.910	0.935	0.909

Note. T-size CFI is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: 4F-2nd
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: 4F-A
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.884 < fair < 0.939 < close** for model: 4F-B
Note. The T-size equivalents of the conventional CFI cut-off values (poor < 0.90 < fair < 0.95 < close) are **poor < 0.885 < fair < 0.94 < close** for model: 4F-bifactor

The comparative fit indices reveal systematic patterns across models. **CFI values:** 4F-bifactor (0.935) > 4F-B (0.910) $\approx$ 4F-2nd (0.909) > 4F-A (0.887), with T-size CFI showing identical rankings (0.919, 0.892, 0.890, 0.867). **TLI values:** 4F-bifactor (0.916) > 4F-B (0.898) = 4F-2nd (0.898) > 4F-A (0.873). **RMSEA values:** 4F-bifactor (0.053, 90% CI [0.049, 0.057]) < 4F-B (0.058) $\approx$ 4F-2nd (0.059) < 4F-A (0.065), with T-size RMSEA showing consistent patterns. **SRMR values:** 4F-bifactor (0.041) < 4F-B (0.051) $\approx$ 4F-2nd (0.051) < 4F-A (0.054). These indices consistently favor the bifactor model, followed by 4F-B and 4F-2nd (nearly tied), with 4F-A showing the poorest fit. However, **critical caution is warranted:** as discussed in Appendix Table 3, traditional cutoff values (CFI > .95, RMSEA < .06) should not be applied mechanically. Bifactor models are structurally more complex (additional general factor with 23 loadings), inherently producing better fit indices due to additional parameters, which can artificially inflate fit without necessarily reflecting superior theoretical validity. **Notably, ALL models fail to achieve "good fit" (CFI > 0.95)** except the bifactor model (CFI = 0.935), which still falls short of this benchmark. The fit improvements represent relatively small effect size differences: $\Delta CFI = 0.025$ between 4F-B and bifactor, $\Delta CFI = 0.023$ between 4F-A and 4F-B. **T-size indices** (T-size CFI, T-size RMSEA) adjust for model complexity and sample size, providing alternative perspectives that still favor the bifactor model but with reduced magnitude. For example, T-size CFI for bifactor (0.919) shows smaller advantage over 4F-B (0.892, $\Delta = 0.027$) than conventional CFI suggests ($\Delta = 0.025$). The near-identical fit between 4F-B (CFI = 0.910) and 4F-2nd (CFI = 0.909) suggests these models explain covariance patterns equivalently, with 4F-2nd offering greater parsimony (50 parameters) compared to 4F-B (53 parameters). **Model selection should prioritize theoretical interpretability, practical utility, and parameter stability** over marginal fit improvements. Given the minimal fit differences between 4F-B and 4F-2nd, information criteria (AIC, BIC) that penalize model complexity will be critical for selecting between these models.

Information criteria

	4F-A	4F-B	4F-bifactor	4F-2nd
Log-likelihood	-29694.806	-28651.621	-28531.297	-28660.484
Number of free parameters	55.000	53.000	79.000	50.000
Akaike (AIC)	59499.611	57409.242	57220.593	57420.969
Bayesian (BIC)	59772.064	57671.787	57611.935	57668.653
Sample-size adjusted Bayesian (SSABIC)	59597.376	57503.452	57361.019	57509.846

Information criteria provide model comparison metrics that balance fit against model complexity, with lower values indicating better models. **AIC values:** 4F-bifactor (57220.593) < 4F-B (57409.242) < 4F-2nd (57420.969) < 4F-A (59499.611). **BIC values:** 4F-2nd (57668.653) < 4F-B (57671.787) < 4F-bifactor (57611.935) < 4F-A (59772.064). **Importantly, AIC and BIC provide contradictory rankings for the top three models.** AIC favors the bifactor model (most complex, 79 parameters), while BIC favors the second-order model (most parsimonious, 50 parameters). This discrepancy reflects the different penalties these criteria apply for model complexity: BIC imposes a stronger penalty for additional parameters, particularly in large samples ($n = 1047$), favoring parsimony. **Critical interpretation principle: Information criteria are ONLY meaningful for relative comparison, not absolute evaluation.** The absolute values (e.g., AIC = 57220.593) provide no information about model quality - only the **differences** between models matter. **AIC differences:** By this criterion, 4F-B and 4F-2nd receive "essentially no support" relative to bifactor, though both are dramatically better than 4F-A ($\Delta AIC > 2000$). **BIC differences:** 4F-2nd is the reference (lowest). 4F-B is $\Delta BIC = 3.134$ higher (essentially equivalent), 4F-bifactor is $\Delta BIC = -56.718$ lower (actually favors bifactor despite stronger parsimony penalty!), and 4F-A is $\Delta BIC = 2103.411$ higher (dramatically worse). The small $\Delta BIC = 3.134$ between 4F-2nd and 4F-B suggests these models are **essentially equivalent** in terms of fit-parsimony balance. **Interpretation:** If parsimony is highly valued, BIC supports the second-order model. If fit maximization is prioritized, AIC supports the bifactor model. The near-equivalence between 4F-B and 4F-2nd ($\Delta BIC = 3.134$) suggests researchers could reasonably select either based on theoretical considerations. The original 4F-A model is clearly inferior by all criteria, supporting Q5P removal.

Metric	4F-A	4F-B	4F-bifactor	4F-2nd
Root mean square error of approximation (RMSEA)	0.065	0.058	0.053	0.059
RMSEA 90% CI lower bound	0.062	0.055	0.049	0.055
RMSEA 90% CI upper bound	0.069	0.062	0.057	0.062
RMSEA p-value	3.111×10^{-13}	7.256×10^{-5}	0.092	4.994×10^{-5}
T-size RMSEA	0.069	0.062	0.057	0.062
Standardized root mean square residual (SRMR)	0.054	0.051	0.041	0.051
Hoelter's critical N ($\alpha = .05$)	222.694	266.755	311.093	265.477
Hoelter's critical N ($\alpha = .01$)	235.960	283.401	331.719	281.936
Goodness of fit index (GFI)	0.898	0.916	0.937	0.914
McDonald fit index (MFI)	0.594	0.684	0.757	0.679
Expected cross validation index (ECVI)	1.379	1.075	0.895	1.086

Note. T-size RMSEA is computed for $\alpha = 0.05$
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.087 < poor** for model: 4F-bifactor
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: 4F-2nd
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: 4F-A
Note. The T-size equivalents of the conventional RMSEA cut-off values (close < 0.05 < fair < 0.08 < poor) are **close < 0.055 < fair < 0.088 < poor** for model: 4F-B

The detailed RMSEA metrics reinforce the fit patterns. The bifactor model demonstrates the closest fit (RMSEA = 0.053, 90% CI [0.049, 0.057]), approaching the "close fit" criterion (< 0.05), with the RMSEA p-value = 0.092 testing the null hypothesis that $RMSEA \leq 0.05$. The p-value = 0.092 (not statistically significant) suggests the data do not reject the hypothesis of close fit, though this test lacks power. The T-size RMSEA (0.057) adjusts for sample size and model complexity, remaining below 0.06. Models 4F-B and 4F-2nd show acceptable fit (RMSEA = 0.058 and 0.059, respectively), with 90% CI upper bounds at 0.062, marginally exceeding the "close fit" threshold but well below "poor fit" (> 0.08). The T-size RMSEA for both models (0.062) adjusts upward slightly, indicating that after accounting for model complexity, these models' fit is marginally worse than conventional RMSEA suggests. The original 4F-A model shows adequate but weaker fit (RMSEA = 0.065, T-size RMSEA = 0.069, 90% CI upper bound = 0.069). **Hoelter's critical N values** indicate the sample size at which the chi-square test would not reject the model at $\alpha = .05$ and $\alpha = .01$. The bifactor model shows the highest critical N ($.05 = 311.093$, $.01 = 331.719$), indicating it would maintain acceptable fit in much smaller samples, while 4F-A shows the lowest (222.694, 235.960), suggesting it requires the current large sample to avoid rejection. The consistent pattern across all RMSEA metrics (conventional, T-size, confidence intervals, Hoelter's N) supports the ranking: bifactor > 4F-B $\approx$ 4F-2nd > 4F-A, though the differences between top models are relatively small ($\Delta RMSEA = 0.005$ - 0.006), representing minor effect size improvements.

R-Squared

	R²			
	4F-A	4F-B	4F-bifactor	4F-2nd
Q3_F	0.196	0.194	0.574	0.195
Q4_F	0.148	0.146	0.440	0.146
Q10_F	0.543	0.538	0.538	0.538
Q15_F	0.243	0.243	0.341	0.244
Q16_F	0.284	0.282	0.281	0.283
Q17_F	0.776	0.779	0.792	0.778
Q18_F	0.723	0.727	0.744	0.727
Q5_P	0.413			
Q6_P	0.407	0.386	0.393	0.392
Q7_P	0.403	0.410	0.412	0.413
Q11_P	0.363	0.364	0.429	0.367
Q19_P	0.674	0.690	0.696	0.695
Q26_P	0.345	0.344	0.350	0.348
Q20_S	0.614	0.616	0.572	0.621
Q21_S	0.248	0.248	0.238	0.253
Q22_S	0.515	0.514	0.600	0.504
Q8_A	0.211	0.210	0.205	0.211
Q9_A	0.321	0.320	0.312	0.322
Q12_A	0.456	0.455	0.462	0.457
Q13_A	0.369	0.373	0.377	0.373
Q14_A	0.464	0.454	0.449	0.454
Q23_A	0.273	0.276	0.277	0.272
Q24_A	0.236	0.242	0.266	0.240
Q25_A	0.235	0.240	0.247	0.239
psycho				1.000
physical				0.812
social				0.532
environment				0.433

The R² comparative table reveals how different model structures partition variance in observed indicators. **Item-level patterns:** Most items show relatively stable R² across models 4F-A, 4F-B, and 4F-2nd (e.g., Q17F ranges 0.776-0.779, Q18F ranges 0.723-0.727), as expected since these models specify similar first-order factor structures with primary differences in factor intercorrelations or hierarchical structure. The bifactor model shows markedly different R² values due to its dual-loading structure, with some items showing substantial increases (Q3F: 0.196 in 4F-A → 0.574 in bifactor, +0.378; Q4F: 0.148 → 0.440, +0.292; Q22S: 0.515 → 0.600, +0.085) and others showing modest increases or even decreases (Q8A: 0.211 → 0.205, -0.006; Q21S: 0.248 → 0.238, -0.010). **Absence of Q5P:** The 4F-A model includes Q5P (R² = 0.413), while all other models exclude it, making direct comparison incomplete. Q5P's moderate R² (0.413) suggests it was an adequate psychological indicator in terms of variance explained, though its cross-loadings on other domains compromised construct validity.

First-order factor R² in 4F-2nd: The second-order model shows R² for first-order factors (psycho = 1.000 due to identification constraint, physical = 0.812, social = 0.532, environment = 0.433), indicating variance explained by the higher-order QOL factor. High R² for physical (81.2% explained) justifies the hierarchical structure, while lower R² for social and environment (< 55%) suggest these domains retain substantial unique information beyond general quality of life. **Interpretation caution:** Higher R² values in the bifactor model do not automatically indicate superior measurement—they reflect the additional general factor capturing common variance across all items. The dramatic R² increases for weak indicators in the four-factor model (Q3F, Q4F) suggest these items primarily measure general wellbeing rather than specific physical quality of life, supporting the bifactor conceptualization where items load on both general and specific factors. However, this also raises the question of whether these items should be retained, revised, or replaced to better capture specific physical quality of life facets.

Comprehensive model comparison and selection guidance:

The four models represent theoretically distinct structures for quality of life measurement, each with different assumptions and implications:

- (1) **4F-A (original four correlated factors with Q5P)**: Assumes distinct but related domains. Shows poorest fit (CFI = 0.887, RMSEA = 0.065), extensive cross-loadings (MI up to 96.566), and is decisively rejected by all criteria (AIC = 59499.611, BIC = 59772.064, both dramatically higher than alternatives). **This model is not recommended.**
- (2) **4F-B (four factors without Q5P)**: Addresses discriminant validity concerns by removing problematic item. Shows meaningful fit improvement (CFI = 0.910, RMSEA = 0.058) and dramatically improved information criteria (Δ AIC = -2090.369, Δ BIC = -2100.277 relative to 4F-A). However, extremely high physical-psycho correlation ($r = 0.914$) persists, questioning empirical distinctiveness of these domains. Reliability remains adequate ($\omega_{\text{psycho}} = 0.796$ despite losing Q5P). This model is **theoretically straightforward and empirically acceptable**, suitable when domain-specific scores are needed and Q5P's cross-loading nature is problematic for the research question.
- (3) **4F-2nd (second-order with Q5P removed)**: Posits domains as manifestations of general QOL. Shows fit nearly identical to 4F-B (CFI = 0.909, RMSEA = 0.059) with greater parsimony (50 vs. 53 parameters). **BIC favors this model** (lowest BIC = 57668.653, though only Δ BIC = 3.134 better than 4F-B, essentially equivalent). Provides clear hierarchical structure where physical ($R^2 = 81.2\%$) is strongly determined by general QOL, while social ($R^2 = 53.2\%$) and environment ($R^2 = 43.3\%$) retain meaningful domain-specific variance. This model is **theoretically elegant and parsimonious**, suitable when general quality of life is the primary construct of interest but domain-specific scores also provide interpretable unique information.
- (4) **4F-bifactor (with Q5P removed)**: Allows items to load on both general QOL and specific domains simultaneously. Shows best fit by conventional indices (CFI = 0.935, RMSEA = 0.053, SRMR = 0.041) and **AIC favors this model** (lowest AIC = 57220.593, though Δ AIC = 188.649-200.376 better than 4F-B/4F-2nd). However, model is complex (79 parameters) and prone to overfitting. **Critical limitation**: JASP does not provide bifactor-specific reliability indices (ω_H , ω_{HS} , ECV) needed to evaluate whether specific factors provide meaningful reliable variance beyond the general factor. Without these metrics, interpretation remains incomplete. This model is **statistically optimal but theoretically complex**, suitable only if bifactor reliability analyses (conducted in lavaan/R) confirm specific factors retain meaningful reliable variance and substantive interpretation justifies the added complexity.

Critical considerations for model selection:

- (1) **Theoretical coherence**: Do the chosen structure and assumptions align with quality of life theory and prior research? Systematic reviews most strongly support the four-correlated-factor structure, though recent studies introduce bifactor and second-order alternatives.
- (2) **Practical utility**: Will domain-specific scores be used? If yes, how will they be interpreted?
- If only total score needed → 4F-2nd or 4F-bifactor (if ω_{HS} analyses support specific factors)
 - If domain profiles critical → 4F-B (retains interpretable domain scores)
 - If both general and domain scores needed with clear hierarchy → 4F-2nd
- (3) **Discriminant validity**: The persistently high physical-psycho correlation ($r > .90$ in 4F-B) questions whether these domains are empirically distinguishable. Hierarchical structures (4F-2nd, 4F-bifactor) explicitly acknowledge this commonality through general/higher-order factors rather than pretending domains are distinct when data suggest otherwise.
- (4) **Parsimony**: Simpler models (4F-B: 53 parameters, 4F-2nd: 50 parameters) are easier to interpret, more likely to replicate across samples, and less prone to overfitting compared to complex models (4F-bifactor: 79 parameters). BIC's preference for 4F-2nd reflects this principle.
- (5) **Information criteria interpretation**: AIC and BIC provide contradictory recommendations (AIC favors bifactor, BIC favors second-order), reflecting different philosophies about the fit-parsimony trade-off. In large samples ($n = 1047$), BIC's stronger parsimony penalty may be more appropriate to avoid overparameterization. The near-equivalence between 4F-B and 4F-2nd (Δ BIC = 3.134, Δ CFI = 0.001) suggests **researchers could reasonably choose either** based on theoretical considerations rather than statistical criteria alone.
- (6) **Q5P decision**: Removing Q5P improves fit (Δ CFI = +0.023), reduces cross-loadings (maximum MI: 96.566 → 66.643), and improves discriminant validity (psycho-environment r : 0.702 → 0.656), with minimal reliability cost ($\Delta\omega = -0.027$). **Recommendation: Remove Q5P** unless complete domain coverage is theoretically essential for the specific research application.

Final recommendation:

For most applications, the second-order model (4F-2nd) is recommended as it offers:

- Acceptable fit (CFI = 0.909, RMSEA = 0.059) nearly identical to 4F-B
- Greatest parsimony (50 parameters, BIC-preferred)
- Clear hierarchical interpretation (general QOL + domain-specific variances)
- Explicit acknowledgment of high factor intercorrelations through higher-order structure
- Stable parameter estimates likely to replicate across samples

The bifactor model should only be selected if:

- Bifactor reliability analyses (ω_H , ω_{HS} , ECV in lavaan/R) confirm specific factors provide meaningful reliable variance
- Theoretical justification exists for simultaneous general + specific loadings
- Research question requires disentangling general wellbeing from domain-specific effects
- Researchers are prepared to interpret and defend the complex dual-loading structure

Researchers should report multiple models (at minimum 4F-B and 4F-2nd, potentially 4F-bifactor if reliability analyses support it) to demonstrate robustness of substantive conclusions across model specifications. Transparency about model selection criteria (theoretical, fit indices, information criteria, practical utility) is essential for reproducible science.

4F-B

4F-bifactor

4F-2nd

Covariance tables

4F-A

4F-B

4F-bifactor

4F-2nd

